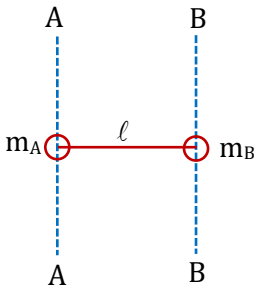
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Moment of Inertia

1. A point mass m_A is connected to a point mass m_B by a massless rod of length ℓ as shown in the figure. It is observed that the ratio of the moment of inertia of the system about the two axes BB and AA, which is parallel to each other and perpendicular to the rod is $\frac{I_{BB}}{I_{AA}} = 3$. The distance of the centre of mass of the system from the point mass m_A is



(A) $(3/4)\ \ell$ (B) $(2/3)\ \ell$

(C) $(1/2)\ \ell$ (D) $(1/4)\ \ell$

PRM091

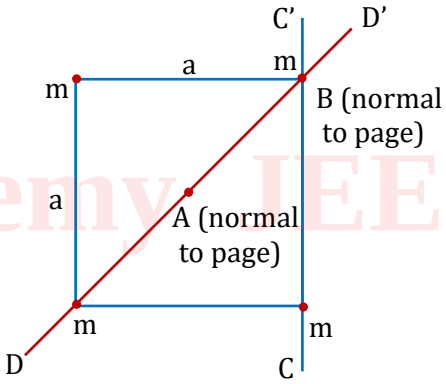
2. The square structure shown in figure consists of four point masses connected by rods of negligible mass. Mark the **CORRECT** option regarding the moment of inertia of the structure about the following axes:

(A) Axis A, passing through the centre of the structure and normal to its plane $I_A = ma^2$

(B) Axis B, passing through one of the point masses and normal to the plane of the structure $I_B = 3\ ma^2$

(C) Axis CC', passing through two adjacent point masses $I_C = 2\ ma^2$ and

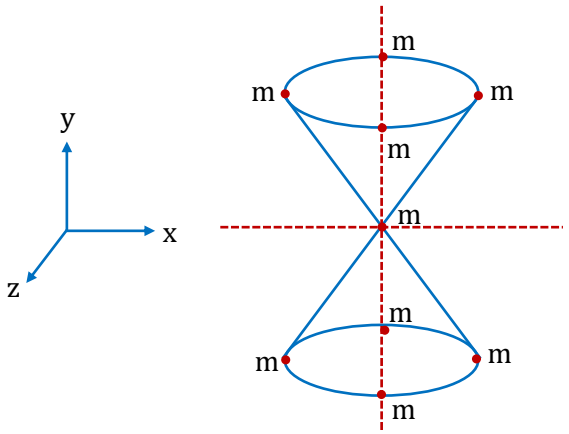
(D) Axis DD', along the diagonal of the structure $I_D = 2\ ma^2$



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PRM092

3. Nine point masses of mass m are arranged in a double-cone system as shown in diagram. Radius of each cone is R and height of each cone is H. The moment of inertia about the x-axis passing through the common vertex is:



(A) $8\ mH^2$ (B) $8\ m(H^2 + R^2)$ (C) $8\ mH^2 + 4\ mR^2$ (D) None of these

PRM093

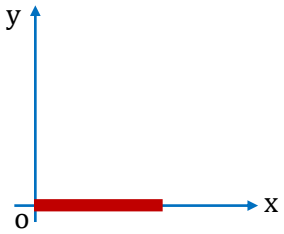
For More Join: @IITJEE_Advanced

4. A rigid body can be hinged about any point on the x-axis. When it is hinged such that the hinge is at x, the moment of inertia is given by $I = 2x^2 - 12x + 27$. The x-coordinate of centre of mass is

(A) $x = 2$ (B) $x = 0$ (C) $x = 1$ (D) $x = 3$

PRM094

5. The figure shows a uniform rod lying along the x-axis. The locus of all the points lying on the xy-plane, about which the moment of inertia of the rod is same as that about O is :-



(A) an ellipse (B) *a* circle (C) *a* parabola (D) *a* straight line

PRM097

6. A thin uniform rod of mass M and length L has its moment of inertia I_1 about its perpendicular bisector. The rod is bend in the form of a semi-circular arc. Now its moment of inertia through the centre of the semi-circular arc and perpendicular to its plane is I_2. The ratio of $I_1 : I_2$ will be _____.

(A) < 1 (B) > 1 (C) = 1 (D) can't be said

PRM098

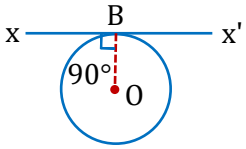
7. A circular disc X of radius R is made from an iron plate of thickness t and another disc Y of radius $4R$ is made from an iron plate of thickness $t/4$. Then the relation between the moment of inertia I_X and I_Y is

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(A) $I_Y = 32\, I_X$ (B) $I_Y = 16\, I_X$ (C) $I_Y = I_X$ (D) $I_Y = 64\, I_X$

PRM099

8. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' is



(A) $\rho L^3/8\pi^2$ (B) $\rho L^3/16\pi^2$ (C) $5\,\rho L^3/16\pi^2$ (D) $3\,\rho L^3/8\pi^2$

PRM100

9. One solid sphere A and another hollow sphere B are of same mass and same outer radii. Their moment of inertia about their diameters are respectively I_A and I_B such that-
(Where d_A and d_B are their densities)

(A) $I_A = I_B$ (B) $I_A > I_B$ (C) $I_A < I_B$ (D) $\frac{I_A}{I_B} = \frac{d_A}{d_B}$

PRM101

10. Three bodies have equal mass m. Body A is solid cylinder of radius R, body B is a square lamina of side R, and body C is a solid sphere of radius R. Which body has the smallest moment of inertia about an axis passing through their centre of mass and perpendicular to the plane (in case of lamina)

(A) A (B) B (C) C (D) A and C both

PRM102

For More Join: @IITJEE_Advanced

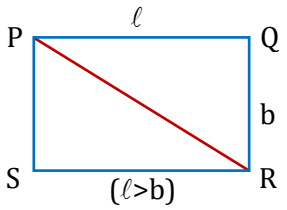
11. For the same total mass which of the following will have the largest moment of inertia about an axis passing through its centre of mass and perpendicular to the plane of the body

(A) a disc of radius a (B) a ring of radius a

(C) a square lamina of side $2a$ (D) four rods forming a square of side $2a$

PRM103

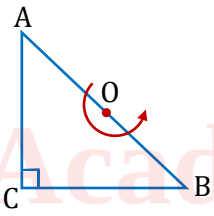
12. Moment of inertia of a rectangular plate about an axis passing through P and perpendicular to the plate is I. Then moment of PQR about an axis perpendicular to the plane of the plate:



(A) about $P = I/2$ (B) about $R = I/2$ (C) about $P > I/2$ (D) about $R > I/2$

PRM105

13. Find the moment of inertia of a plate cut in shape of a right angled triangle of mass M, about an axis perpendicular to the plane of the plate and passing through the mid point of side AB.
(Side $AC = BC = a$)

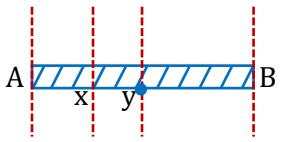


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(A) $\frac{Ma^2}{12}$ (B) $\frac{Ma^2}{6}$ (C) $\frac{Ma^2}{3}$ (D) $\frac{2Ma^2}{3}$

PRM110

14. A rod of non uniform mass distribution in which the mass increases from left end (A) to right end (B). It is easiest to rotate the rod about an axis

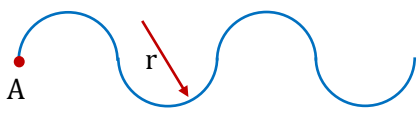


(A) Passing through A (B) Passing through B

(C) Passing through x (D) Passing through y

PRM242

15. A uniform rod of mass $2M$ is bent into four adjacent semicircles each of radius r, all lying in the same plane. The moment of inertia of the bent rod about an axis through one end A and perpendicular to plane of rod is

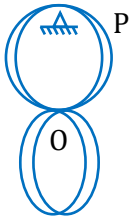


(A) 22 Mr^2 (B) 88 Mr^2 (C) 44 Mr^2 (D) 66 Mr^2

PRM244

For More Join: @IITJEE_Advanced

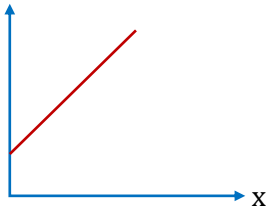
16. Two identical rings each of mass m with their planes mutually perpendicular, radius R are welded at their point of contact O. If the system is free to rotate about an axis passing through the point P perpendicular to the plane of the paper the moment of inertia of the system about this axis is equal to

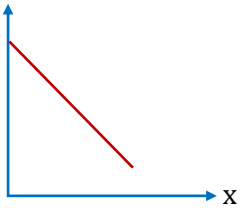


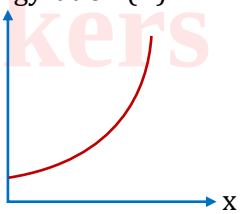
(A) 6.5 mR^2 (B) 12 mR^2 (C) 6 mR^2 (D) 11.5 mR^2

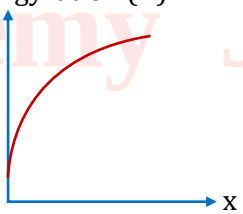
PRM245

17. A uniform disc has radius R. If a concentric smaller disc of radius x is removed from the bigger disc then radius of gyration of the remaining disc about an axis passing through centre and perpendicular to plane of disc varies with x, according to the graph:

(A)


(B)


(C)


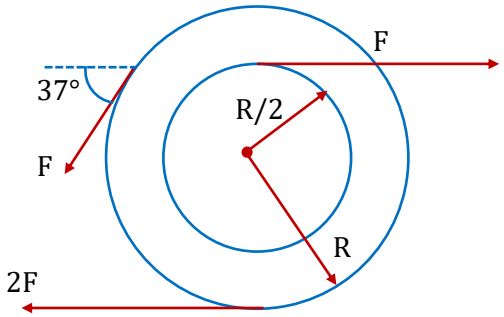
(D)


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PRM246

Torque & Condition of Equilibrium

18. On a disc of radius R a concentric circle of radius $R/2$ is drawn. The disc is free to rotate about a frictionless axis through its centre and perpendicular to plane of the disc. All the three forces (in plane of the disc) shown in figure are act tangentially to their respective circular periphery. The magnitude of the net torque (about centre of disc) acting on the disc is :-



(A) 1.9 FR (B) 2.3 FR (C) 2.5 FR (D) 1.5 FR

PRM241

For More Join: @IITJEE_Advanced

19. A weightless rod is acted on by upward parallel forces of $2N$ and $4N$ at ends A and B respectively. The total length of the rod is $AB = 3m$. To keep the rod in equilibrium a force of $6N$ should act in the following manner:

(A) Downwards at any point between A and B.

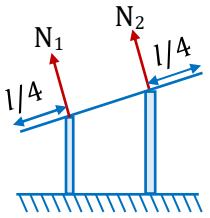
(B) Downwards at mid point of AB.

(C) Downwards at a point C such that $AC = 1m$.

(D) Downwards at a point D such that $BD = 1m$.

PRM120

20. A uniform rod of length l is placed symmetrically on two walls as shown in figure. The rod is in equilibrium. If N_1 and N_2 are the normal forces exerted by the walls on the rod then :-



(A) $N_1 > N_2$

(B) $N_1 < N_2$

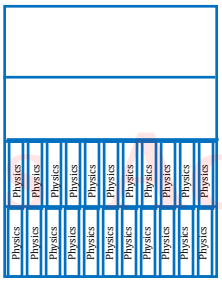
(C) $N_1 = N_2$

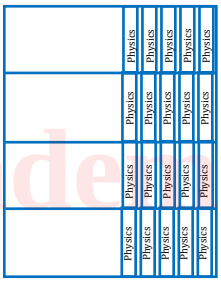
(D) N_1 and N_2 would be in the vertical directions

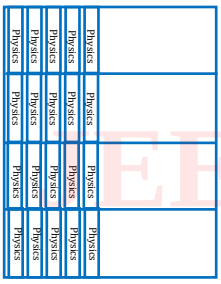
PRM122

21. Same number of books are placed in four book cases as shown. Which bookcase is most likely to topple forward if pulled a little at the top towards right :

(A) 

(B) 

(C) 

(D) 

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PRM126

22. A homogeneous cubical brick lies motionless on a rough inclined surface. The half of the brick which applies greater pressure on the plane is :



(A) left half

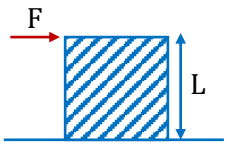
(B) right half

(C) both applies equal pressure

(D) the answer depend upon coefficient of friction

PRM127

23. A cubical block of side L rests on a rough horizontal surface with coefficient of friction μ. A horizontal force F is applied on the block as shown. If the coefficient of friction is sufficiently high so that the block does not slide before toppling, the minimum force required to topple the block is

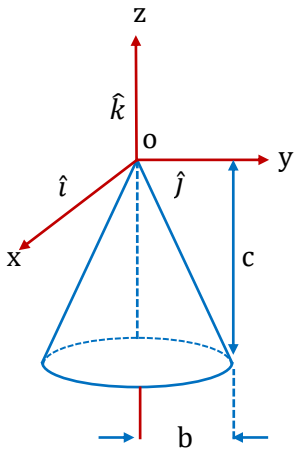


(A) Infinitesimal (B) $mg/4$ (C) $mg/2$ (D) $mg(1 - m)$

PRM128

For More Join: @IITJEE_Advanced

24. A solid cone hangs from a frictionless pivot at the origin O, as shown. If $\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ are unit vectors, and a, b, and c are positive constants, which of the following forces F applied to the rim of the cone at a point P results in a torque τ on the cone with a negative component τ_z?



(A) $F = a\hat{k}$, P is $(0, b, -c)$
(B) $F = -a\hat{k}$, P is $(0, -b, -c)$
(C) $F = a\,\hat{\jmath}$, P is $(-b, 0, -c)$
(D) None

PRM124

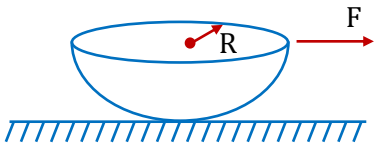
25. A homogeneous disc with a radius 0.2 m and mass 5 kg rotates around an axis passing through its centre. The angular velocity of the rotation of the disc as a function of time is given by the formula $\omega = 2 + 6t$. The tangential force applied to the rim of the disc is

(A) 1 N (B) 2 N (C) 3 N (D) 4 N

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PRM248

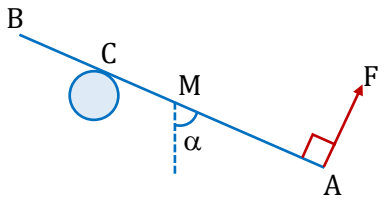
26. A thin uniform hemispherical bowl of mass m and radius R is lying on a smooth horizontal surface. A horizontal force F is now applied perpendicular to the rim of the bowl (see figure). The instantaneous angular acceleration of the bowl will be :-



(A) $\frac{20}{3}\frac{F}{MR}$ (B) $\frac{10}{3}\frac{F}{MR}$ (C) $\frac{40}{3}\frac{F}{MR}$ (D) $\frac{6}{5}\frac{F}{MR}$

PRM249

27. A uniform rod AB of weight 100 N rests on a rough peg at C and a force F acts at A as shown in figure. $BC = CM$, M is the midpoint of AB and $\tan\alpha = 4/3$. The minimum coefficient of friction at C is

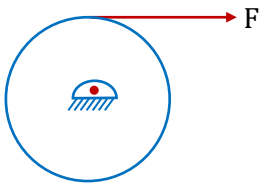


(A) 9/8 (B) 3/2 (C) 8/7 (D) 4/3

PRM250

For More Join: @IITJEE_Advanced

28. A pulley is hinged at the centre and a massless thread is wrapped around it. The thread is pulled with a constant force F starting from rest. As time increases,



(A) its angular velocity increases, but force on hinge remains constant

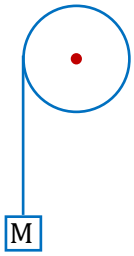
(B) its angular velocity remains same, but force on hinge increases

(C) its angular velocity increases and force on hinge increases

(D) its angular velocity remains same and force on hinge is constant

PRM152

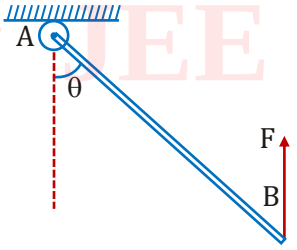
29. A fixed pulley of radius 20 cm and moment of inertia 0.32 $kg.m^2$ about its axle has a massless cord wrapped around its rim. A mass M of 2 kg is attached to the end of the cord. The pulley can rotate about its axis without any friction. The acceleration of the mass M is :- (Assume g = 10 m/s^2)



(A) 1 m/s^2 (B) 3 m/s^2 (C) 2 m/s^2 (D) 4 m/s^2

PRM235

30. As shown in the figure, a uniform rod can be rotated freely about point A in the vertical plane. When a vertical upward force F lifts up the rod slowly from position shown, which of the following statements is correct till the rod becomes horizontal ?



(A) The pulling force is unchanged, its moment of force increases.

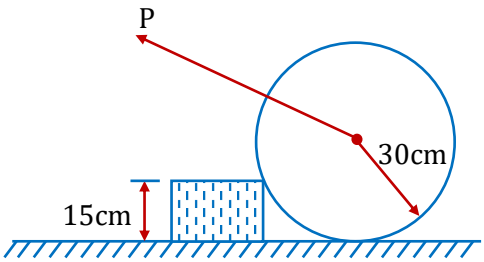
(B) The pulling force is unchanged, its moment of force decreases.

(C) The pulling force increases, its moment of force decreases.

(D) The pulling force decreases, its moment of force decreases.

PRM237

31. In the given figure a uniform wheel of radius 30cm rests against a rigid rectangular block 15cm high. All the surfaces are smooth. The wheel weighs 1000 N. The minimum pull P through the center which will turn the wheel over the block is :-



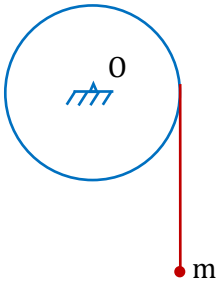
(A) $500\sqrt{3}$ N (B) $1000\sqrt{3}$ N (C) 1000 N (D) $400\sqrt{3}$ N

PRM238

For More Join: @IITJEE_Advanced

Kinetic Energy in Rotation

32. A uniform disc of mass M and radius R is smoothly pivoted at 'O'. A light inextensible string is wrapped over the disc hangs a particle of mass m. If the system is released from rest, assuming string does not slide, what will be the angular speed of the disc as a function of time.



(A) $\omega = \dfrac{2mgt}{(M+2m)R}$ (B) $\omega = \dfrac{4mgt}{(M+2m)R}$

(C) $\omega = \dfrac{mgt}{(M+2m)R}$ (D) $\omega = \dfrac{mgt}{(M+m)R}$

PRM240

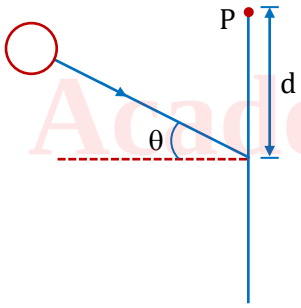
Fixed Axis Angular Momentum & its Conservation

33. A particle of mass 2 kg located at the position $(\hat{\imath}+\hat{\jmath})$ m has a velocity 2 $(+\hat{\imath}-\hat{\jmath}+\hat{k})$ m/s. Its angular momentum about z-axis in $kg\text{-}m^2/s$ is:

(A) zero (B) +8 (C) 12 (D) –8

PRM131

34. A ball of mass m moving with velocity v, collide with the wall elastically as shown in the figure. After impact the change in angular momentum about P is:



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(A) 2 mvd (B) 2 mvd cos θ (C) 2 mvd sin θ (D) zero

PRM132

35. Two uniform spheres of mass M have radii R and $2R$. Each sphere is rotating about a fixed axis through a diameter. The rotational kinetic energies of the spheres are identical. What is the ratio of the magnitude of the angular momenta of these spheres? That is, $\dfrac{L_{2R}}{L_R}$ =

(A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$

PRM133

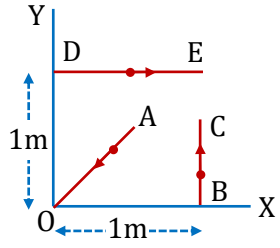
36. A particle of mass m moves with a constant velocity. Which of the following statements is not correct about its angular momentum about point O :

(A) it is zero when it is at A and moving along OA

(B) is the same at all points along the line DE

(C) is of the same magnitude but oppositely directed at B and D

(D) increases as it moves along the line BC



PRM135

For More Join: @IITJEE_Advanced

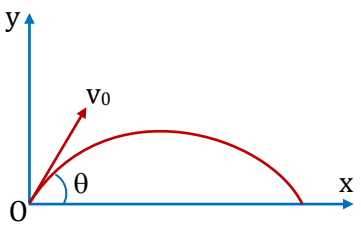
37. A projectile of mass 'm' is thrown from the origin O with a velocity v_0 at an angle θ. Identify the **CORRECT** statement :-

(A) Angular momentum of the particle about O is always zero

(B) Angular momentum of the particle about O increases with time.

(C) A constant torque acts on the particle about O in the clockwise sense.

(D) Angular momentum of the particle is least when it is at the highest position.

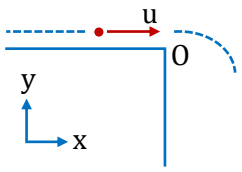


PRM136

38. A particle of mass m is projected from point O with a horizontal velocity 'u' at time t = 0. What is its angular momentum relative to point O as a function of time?

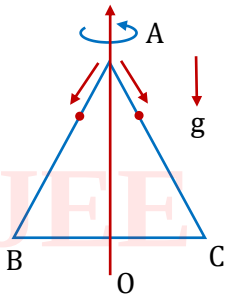
(A) $-\frac{1}{2}mgut^2\hat{k}$ (B) $\frac{1}{2}mgut^2\hat{k}$

(C) $-mgut^2\hat{k}$ (D) $+mgut^2\hat{k}$



PRM138

39. An equilateral triangle ABC formed from a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and the other along AC as shown. Neglecting frictional effects, the quantities that are conserved as the beads slide down, are



Raikers Academy JEE

(A) Angular velocity and total energy (kinetic and potential)

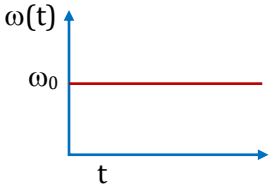
(B) Total angular momentum and total energy

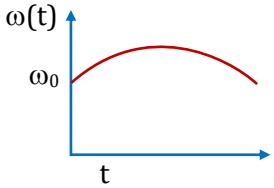
(C) Angular velocity and moment of inertia about the axis of rotation.

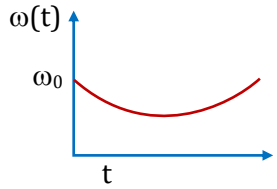
(D) Total angular momentum and moment of inertia about the axis of rotation.

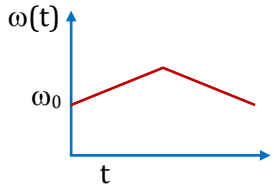
PRM141

40. A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now, the platform is given an angular velocity ω_0. When the tortoise moves along a chord of the platform with a constant velocity (with respect to the platform), the angular velocity of the platform $\omega(t)$. will vary with time t as

(A)


(B)


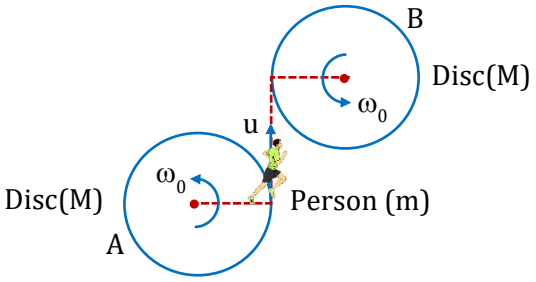
(C)


(D)


PRM142

For More Join: @IITJEE_Advanced

41. Two identical discs each of mass $M(= 4m)$ and radius R are rotating in opposite sense with equal angular speed ω_0 about vertical axes passing through their centres, (as shown in figure). A person of mass m sitting on circumference of disc A jumps with a tangential relative velocity u (after jumping) w.r.t one rotating disc (A) and lands on other disc (B) also tangential. Now the second disc (B) comes to a stop. Find the relative velocity u.



(A) $u = \frac{\omega_0 R}{2}$ (B) $u = \omega_0 R$ (C) $u = \frac{3\omega_0 R}{2}$ (D) $u = 2\omega_0 R$

PRM143

42. A spinning ice skater can increase his rate of rotation by bringing his arms and free leg closer to his body. How does this procedure affect the skater's angular momentum and kinetic energy and what is the work done by the skater?

(A) angular momentum remains the same while kinetic energy increases and work done is positive.

(B) angular momentum remains the same while kinetic energy decreases and work done is negative.

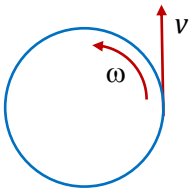
(C) both angular momentum and kinetic energy remain the same and work done is zero.

(D) angular momentum increases while kinetic energy remains the same and work done may be positive or negative.

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PRM145

43. A child with mass m is standing at the edge of a disc with moment of inertia I, radius R, and initial angular velocity ω. See figure given below. The child jumps off the edge of the disc with tangential velocity v with respect to the ground. The new angular velocity of the disc is



(A) $\sqrt{\frac{I\omega^2 - mv^2}{I}}$ (B) $\sqrt{\frac{(I+mR^2)\omega^2 - mv^2}{I}}$

(C) $\frac{I\omega - mvR}{I}$ (D) $\frac{(I+mR^2)\omega - mvR}{I}$

PRM146

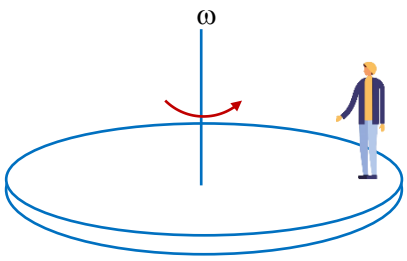
44. A rod of mass M and length ℓ stands along z-axis. Its lower end is hinged at the centre of a disc of same mass M and radius R. The whole arrangement is rotating freely about z-axis with an angular velocity ω_0. The rod falls on disc and rotates with disc. Find the angular speed of the arrangement.

(A) $\frac{\omega_0 R^2}{\ell^2 + R^2}$ (B) $\frac{3\omega_0 R^2}{2\ell^2 + 3R^2}$ (C) $\frac{6\omega_0 R^2}{\ell^2 + 6R^2}$ (D) $\frac{6\omega_0 R}{\ell + 6R}$

PRM147

For More Join: @IITJEE_Advanced

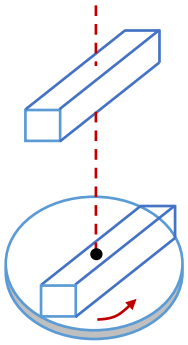
45. A child is standing on the edge of a merry-go-round that has the shape of a disk, as shown in the figure above. The mass of the child is 40 kilograms. The merry-go-round has a mass of 200 kilograms and a radius of 2.5 meters, and it is rotating with an angular velocity of ω = 2.0 radians per second. The child then walks slowly toward the center of the merry-go-round. What will be the final angular velocity of the merry-go-round when the child reaches the center ? (The size of the child can be neglected)



(A) 2.0 rad/s (B) 2.2 rad/s (C) 2.4 rad/s (D) 2.8 rad/s

PRM148

46. A uniform disk turns at 2.4 rev/s around a frictionless axis. A nonrotating rod, of the same mass as the disk and length equal to the disk's diameter is dropped onto the freely spinning disk figure. They then both turn around the axis with their centres superposed. What is the angular frequency in rev/s of the combination.



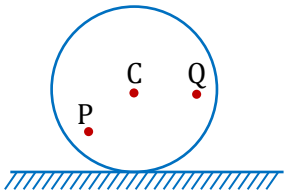
(A) 1.2 rev/s
(B) 2.0 rev/s
(C) 1.44 rev/s
(D) 0.96 rev/s

PRM251

Rankers Academy JEE

Kinematics of Combined Translational & Rotational Motion

47. A disc is rolling without slipping with angular velocity ω. P and Q are two points equidistant from the centre C. The order of magnitude of velocity is :-

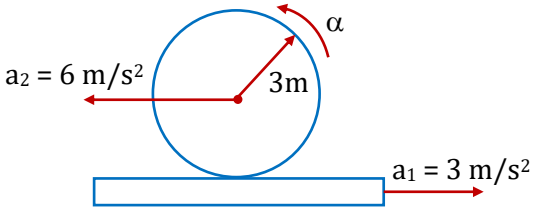


(A) $v_Q > v_C > v_p$ (B) $v_P > v_C > v_Q$ (C) $v_P = v_C$, $v_Q = v_C/2$ (D) $v_P < v_C > v_Q$

PRM159

48. In the following figure, a sphere of radius 3 m rolls on a plank. The accelerations of the sphere and the plank are indicated. The value of α is

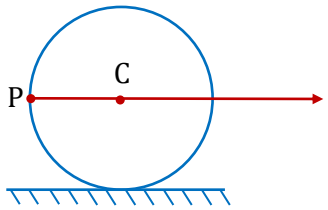
(A) 3 rad/s^2
(B) 6 rad/s^2
(C) 3 rad/s^2 (opposite to the direction shown in figure)
(D) 1 rad/s^2



PRM160

For More Join: @IITJEE_Advanced

49. A disc of radius R is rolling purely on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is

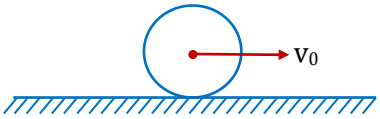


(A) zero (B) 45°
(C) 135° (D) $\tan^{-1}(1/2)$

PRM161

Angular Momentum in Combined Translational & Rotational Motion

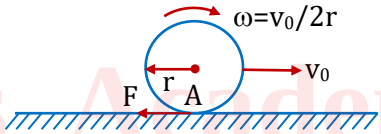
50. A ring of mass m is thrown in horizontal direction on a rough horizontal surface with a speed v_0. The speed of the ring, when it start pure rolling, is –



(A) $\frac{v_0}{2}$ (B) $\frac{v_0}{3}$ (C) $\frac{2v_0}{3}$ (D) $\frac{3v_0}{3}$

PRM192

51. A solid sphere of mass M and radius r slips on a rough horizontal plane. At some instant it has translational velocity v_0 and rotational velocity about the centre $v_0/2r$. The translational velocity after the sphere starts pure rolling

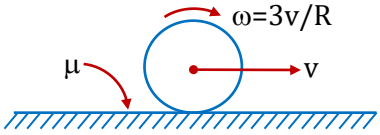


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(A) $6v_0/7$ in forward direction (B) $6v_0/7$ in backward direction
(C) $7v_0/6$ in forward direction (D) $7v_0/6$ in backward direction

PRM193

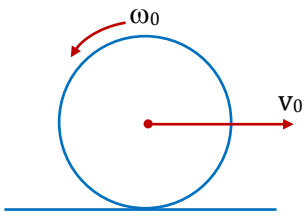
52. A hollow sphere is moving with linear velocity and angular velocity as shown in figure. The time after which pure rolling starts is :



(A) $\frac{2v}{3\mu g}$ (B) $\frac{4v}{7\mu g}$ (C) $\frac{4v}{5\mu g}$ (D) $\frac{5v}{4\mu g}$

PRM194

53. A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is



(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

PRM197

For More Join: @IITJEE_Advanced

54. A hollow sphere of radius R and mass m is fully filled with non viscous liquid of mass m. It is rolled down a horizontal plane such that its centre of mass moves with a velocity v. If it purely rolls

(A) Kinetic energy of the sphere is $\frac{5}{6}mv^2$

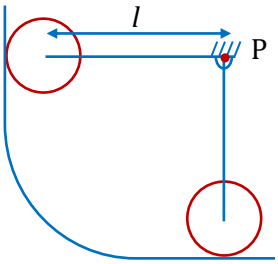
(B) Kinetic energy of the sphere is $\frac{4}{5}mv^2$

(C) Angular momentum of the sphere about a fixed point on ground is $\frac{8}{3}mvR$

(D) Angular momentum of the sphere about a fixed point on ground is $\frac{14}{5}mvR$

PRM172

55. A sphere of mass M and radius R is attached by a light rod of length ℓ to a point P. The sphere rolls without slipping on a circular track as shown. It is released from the horizontal position. The angular momentum of the system about P when the rod becomes vertical is:



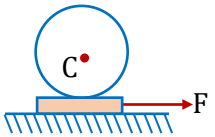
(A) $M\sqrt{\frac{10}{7}gl}[l+R]$ (B) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{2}{5}R\right]$

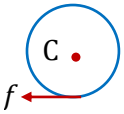
(C) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{7}{5}R\right]$ (D) $M\sqrt{\frac{10}{7}gl}\left[l-\frac{2}{5}R\right]$

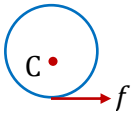
PRM173

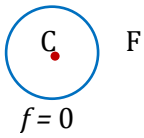
Dynamics of Combined Translational & Rotational Motion

56. A cylinder is placed on a rough plank which in turn is placed on a smooth surface. The plank is pulled with a constant force F. The friction force can be given by which of the following diagrams



(A) 

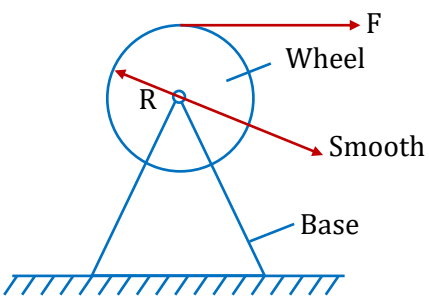
(B) 

(C) 

(D) cannot be interpreted

PRM166

57. A wheel (disc of mass m, radius R) is mounted on a base of mass M. The whole system is placed on rough horizontal surface. The coefficient of friction between the base and the surface is μ. Initially the system is at rest. A time varying horizontal force $F = 2t$ is applied tangentially on the wheel, where t is time in second and force F is in newton. The kinetic energy of the disc when the system is about to slide on ground is :-

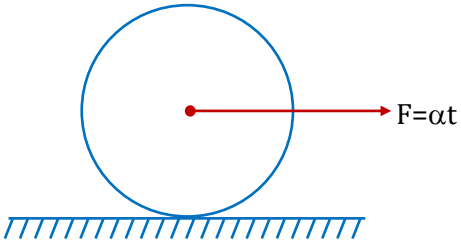


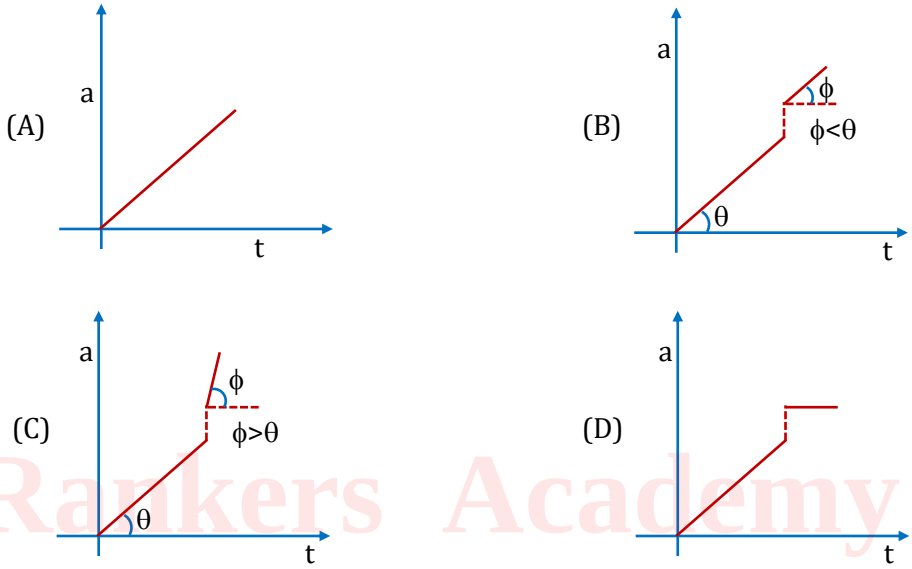
(A) $\frac{1}{m}\left(\frac{\mu(M)g}{2}\right)^4$ (B) $\frac{1}{m}\left(\frac{\mu(m+M)g}{2}\right)^4$ (C) $\frac{1}{m}\left(\frac{\mu(m+M)g}{2}\right)^2$ (D) $\frac{1}{m}\left(\frac{\mu(M)g}{2}\right)^2$

PRM239

For More Join: @IITJEE_Advanced

58. A sphere of mass m and radius R is kept on a rough horizontal surface and pulled with a horizontal force $F = \alpha t$ where α is a constant and t is time. The coefficient of static friction is μ_s and coefficient of kinetic friction is μ_k. Which of the following graph correctly shows the variation of acceleration of sphere with time ?





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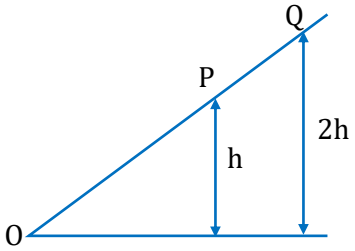
PRM247

59. A body kept on a smooth horizontal surface is pulled by a constant horizontal force applied at the top point of the body. If the body rolls purely on the surface, its shape can be :

(A) thin pipe (B) uniform cylinder (C) uniform sphere (D) thin spherical shell

PRM171

60. A ball rolls down an inclined plane as shown in figure. The ball is first released from rest from P and then later from Q. Which of the following statement is/are correct?



(i) The ball takes twice as much time to roll from Q to O as it does to roll from P to O.

(ii) The acceleration of the ball at Q is twice as large as the acceleration at P.

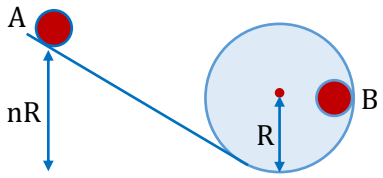
(iii) The ball has twice as much K.E. at O when rolling from Q as it does when rolling from P.

(A) i, ii only (B) ii, iii only (C) i only (D) iii only

PRM187

For More Join: @IITJEE_Advanced

61. A metal ball of mass m is put at the point A of a loop tract and the vertical distance of A from the lower most point of tract is n times the radius R of the circular part. The linear velocity of ball when it rolls to the point B to a height R in the circular tract will be



(A) $\sqrt{\frac{5g(n-1)R}{7}}$ (B) $\sqrt{\frac{10g(n-1)R}{7}}$ (C) $\sqrt{\frac{5g(n+1)R}{7}}$ (D) $\sqrt{\frac{10g(n+1)R}{7}}$

PRM252

62. A solid sphere is under pure rolling on a rough fixed incline plane of angle θ. Choose the correct options if only contact forces and gravity are acting :

(A) Frictional force will be down the incline if sphere rolls up the incline and it will be up the incline if sphere rolls down the incline.

(B) Frictional force will be down the incline whether spheres rolls up the incline or down the incline

(C) Frictional force and acceleration of the body will increase with the increase in angle of incline plane

(D) Velocity and acceleration of the point of contact of sphere with incline will be zero during the motion.

PRM180

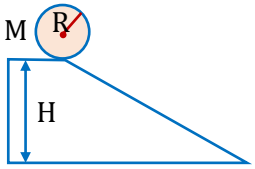
Rankers Academy JEE

63. A solid uniform disk of mass m rolls without slipping down a fixed inclined plane with an acceleration a. The frictional force on the disk due to surface of the plane is :

(A) 2 ma (B) 3/2 ma (C) ma (D) 1/2 ma

PRM181

64. The cylinder shown, with mass M and radius R, has a radially dependent density. The cylinder starts from rest and rolls without slipping down an inclined plane of height H. At the bottom of the plane its translational speed is $(8gH/7)^{1/2}$. Which of the following is the rotational inertia of the cylinder?



(A) $\frac{1}{2}MR^2$ (B) $\frac{3}{4}MR^2$ (C) $\frac{7}{8}MR^2$ (D) MR^2

PRM182

65. A uniform disk of mass m and radius R rolls without slipping down an incline plane of length l and inclination θ. Initially the disk was at rest at the top of the incline plane. Its angular momentum about the point of contact with the inclined plane when it reaches the bottom will be equal to :-

(A) $\sqrt{4m^2R^2gl\sin\theta}$ (B) $\sqrt{3m^2R^2gl\sin\theta}$ (C) $\sqrt{\frac{m^2R^2gl\sin\theta}{4}}$ (D) zero

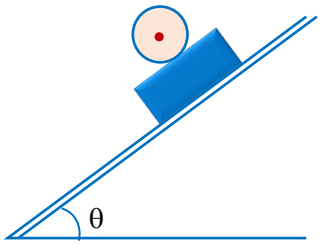
PRM183

For More Join: @IITJEE_Advanced

66. A cylinder rolls up an inclined plane, reaches some height, and then rolls down (without slipping throughout these motions). The directions of the frictional force acting on the cylinder are
(A) Up the incline while ascending and down the incline while descending
(B) Up the incline while ascending as well as descending
(C) down the incline while ascending and up the incline while descending
(D) down the incline while ascending as well as descending

PRM185

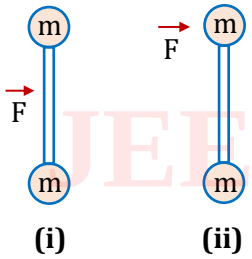
67. A plank of mass M is placed over smooth inclined plane and a sphere is also placed over the plank. Friction is sufficient between sphere and plank. If plank and sphere are released from rest, the frictional force on sphere is:



(A) up the plane (B) down the plane (C) horizontal (D) zero

PRM188

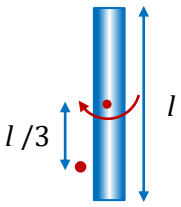
68. A force F is applied to a dumbbell for a time interval, t, first as in (i) and then as in (ii). In which case does the dumbbell acquire the greater centre-of-mass speed?
(A) (i)
(B) (ii)
(C) there is no difference
(D) the answer depends on the rotational inertia of the dumbbell



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PRM204

69. A uniform rod of length l and mass M rotating about a fixed vertical axis on a smooth horizontal table. It elastically strikes a particle placed at a distance $l/3$ from its axis and stops. Mass of the particle is



(A) $3M$ (B) $\frac{3M}{4}$ (C) $\frac{3M}{2}$ (D) $\frac{4M}{3}$

PRM210

70. A mass m is moving at speed v perpendicular to a rod of length d and mass $M = 6m$ which pivots around a frictionless axle running through its centre. It strikes and sticks to the end of the rod. The moment of inertia of the rod about its centre is $Md^2/12$. Then the angular speed of the system right after the collision is
(A) $2v/d$ (B) $2v/(3d)$ (C) v/d (D) $3v/(2d)$

PRM211

For More Join: @IITJEE_Advanced

71. Two particles each of mass M are connected by a massless rod of length l. The rod is lying on the smooth surface. If one of the particle is given an impulse Mv as shown in the figure then angular velocity of the rod would be



(A) v/l (B) $2v/l$ (C) $v/2l$ (D) None

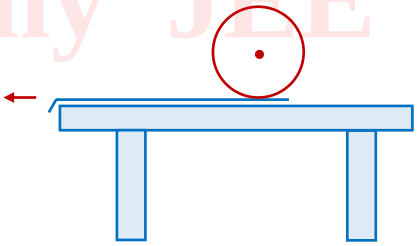
PRM212

72. A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. An impulse J is applied to the end B, perpendicular to the rod in the horizontal direction. Speed of particle P at a distance $\frac{l}{6}$ from the centre towards A of the rod after time $t = \frac{\pi m l}{12J}$ is

(A) $2\frac{J}{m}$ (B) $\frac{J}{\sqrt{2}m}$ (C) $\frac{J}{m}$ (D) $\sqrt{2}\frac{J}{m}$

PRM213

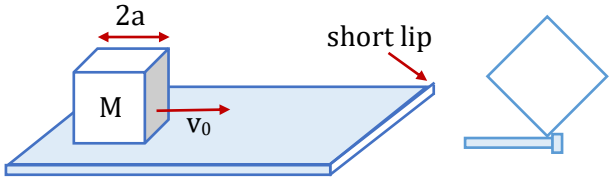
73. A thin table cloth covers a horizontal table and a uniform body of round shape lies on top of it. The table cloth is pulled from under the body, and friction causes the body to slide and rotate. What is the body's final motion on the table?
(Assume that the table is so large that the body does not fall off it.)



(A) Body will finally roll towards left
(B) Body will finally roll towards right
(C) Body will finally come to rest
(D) Any of the above is possible depending on shape of body

PRM174

74. A cube with sides of length $2a$ and mass M is moving with an initial speed v_0 along a frictionless table. When the cube reaches the end of the table it is caught abruptly by a short lip and begins to rotate. The minimum speed v_0 such that the cube falls off the table is :-



(A) $\sqrt{\frac{16ag(\sqrt{2}-1)}{3}}$ (B) $\sqrt{16ag(\sqrt{2}-1)}$ (C) $\sqrt{ag(\sqrt{2}-1)}$ (D) $\sqrt{\frac{ag}{3}(\sqrt{2}-1)}$

PRM236

For More Join: @IITJEE_Advanced

75. A sphere is placed rotating with its centre initially at rest in a corner as shown in figure (a) and (b). Coefficient of friction between all surfaces and the sphere is 1/3. Find the ratio of the frictional force $\frac{f_a}{f_b}$ by ground in situations (a) and (b).



(a) **(b)**

(A) 1 (B) $\frac{9}{10}$ (C) $\frac{10}{9}$ (D) none

PRM224

76. One ice skater of mass m moves with speed $2v$ to the right, while another of the same mass m moves with speed v toward the left, as shown in figure *I*. Their paths are separated by a distance b. At $t = 0$, when they are both at $x = 0$, they grasp a pole of length b and negligible mass. For $t > 0$, consider the system as a rigid body of two masses m separated by distance b, as shown in figure II. Which of the following is the correct formula for the motion after $t = 0$ of the skater initially at $y = b/2$?

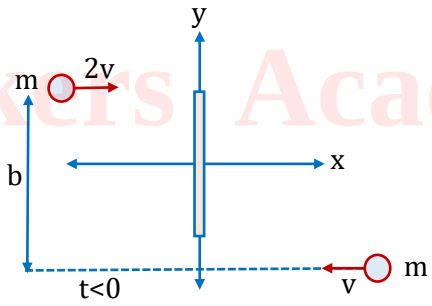


Figure I

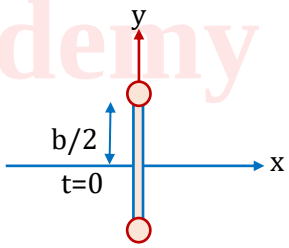


Figure II

(A) $x = 2vt, y = b/2$

(B) $x = vt + 0.5b\sin(3vt/b), y = 0.5b\cos(3vt/b)$

(C) $x = 0.5vt + 0.5b\sin(3vt/b), y = 0.5b\cos(3vt/b)$

(D) $x = 0.5vt + 0.5b\sin(6vt/b), y = 0.5b\cos(6vt/b)$

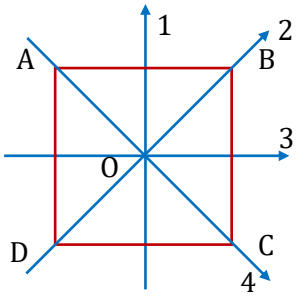
PRM233

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. *ABCD* is a square plate with centre *O*. The moments of inertia of the plate about the perpendicular axis through *O* is *I* and about the axes 1, 2, 3 and 4 are I_1, I_2, I_3 and I_4 respectively. It follows that:



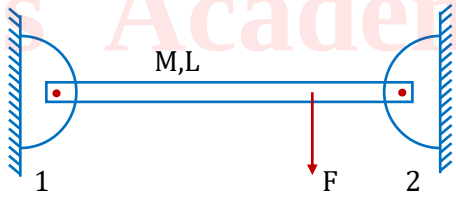
(A) $I_2 = I_3$ (B) $I = I_1 + I_4$

(C) $I = I_2 + I_4$ (D) $I_1 = I_3$

PRM104

2. A uniform rod of mass $M = 2\ kg$ and length L is suspended by two smooth hinges 1 and 2 as shown in the figure. A force $F = 4N$ is applied downward at a distance $L/4$ from the hinge 2. Due to the application of force *F*, the hinge 2 breaks. At this instant, applied force *F* is also removed. The rod starts to rotate downward about hinge 1. ($g = 10\ m/s^2$). Choose the correct statement(s) :-

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(A) The reaction at hinge 1, before hinge 2 breaks, is 11 *N*

(B) The reaction at hinge 1, before hinge 2 breaks, is 24 *N*

(C) The reaction at hinge 1, just after breaking of hinge 2 is 5*N*

(D) The reaction at hinge 1, just after breaking of hinge 2 is 8*N*

PRM253

3. A block with a square base measuring '*a*' × '*a*' and height *h*, is placed on an inclined plane. The coefficient of friction is μ. The angle of inclination (θ) of the plane is gradually increased. The block will

(A) topple before sliding if $\mu > \frac{a}{h}$

(B) topple before sliding if $\mu < \frac{a}{h}$

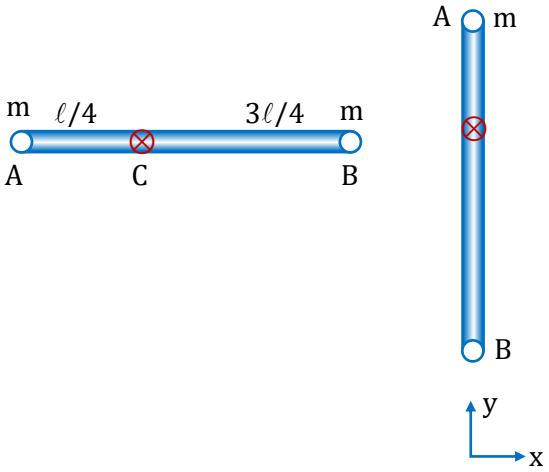
(C) slide before toppling if $\mu > \frac{a}{h}$

(D) slide before toppling if $\mu < \frac{a}{h}$

PRM129

For More Join: @IITJEE_Advanced

4. Two particles each of mass m are attached at end points of a massless rod AB of length ℓ. Rod is hinged at point C as shown. Rod is released from rest from horizontal position. At the instant when rod reaches its vertical position as shown, which of the following is/are **CORRECT** :-



(A) Speed of the particle at B is thrice the speed of particle at A

(B) Net force on particle B is $\frac{6mg}{5}$

(C) Angular acceleration of the system is zero

(D) Both x and y components of hinge force are nonzero

PRM153

5. A rod is connected with the (rigid body) disc at centre of disc point 'P'. The disc can freely rotate about point P where it is hinged with rod and rod can rotate freely about fixed hinge Q. Now whole arrangement is released from horizontal position as shown in the diagram.



(A) Initial angular acceleration of (rigid body) disc is zero.

(B) Initial angular acceleration of (rigid body) disc is $\frac{9g}{8\ell}$.

(C) Angular speed of (rigid body) disc when rod becomes vertical is zero.

(D) Angular speed of (rigid body) disc when rod becomes vertical is $\frac{3}{2}\sqrt{\frac{g}{\ell}}$ rad/s.

PRM154

6. A block of mass m moves on a horizontal rough surface with initial velocity v. The height of the centre of mass of the block is h from the surface. Consider a point A on the surface.

(A) angular momentum about A is mvh initially

(B) the velocity of the block decreases at time passes.

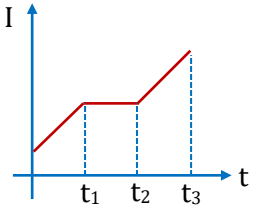
(C) torque of the forces acting on block is zero about A

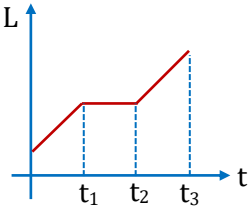
(D) angular momentum is not conserved about A.

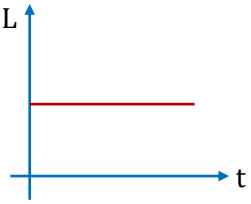
PRM144

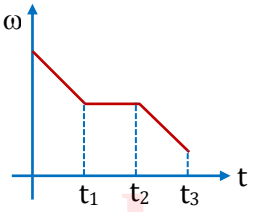
For More Join: @IITJEE_Advanced

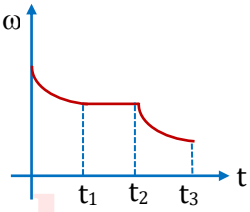
7. A satellite in outer-space is at an initial rotation speed of ω = 4 rad/sec at t = 0. As it deploys its solar panels, it changes its rotational inertia as shown in graph below. The correct plot of magnitude of angular momentum and angular velocity as functions of time :



(A)


(B)


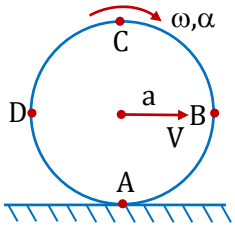
(C)


(D)


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PRM255

8. A circular disc of radius R rolls without slipping on a rough horizontal surface. At the instant shown its linear velocity is V, linear acceleration a, angular velocity ω and angular acceleration α. Four points A, B, C and D lie on its circumference such that the diameter AC is vertical and BD horizontal then choose the correct option(s).



(A) $V_B = \sqrt{V^2 + (R\omega)^2}$

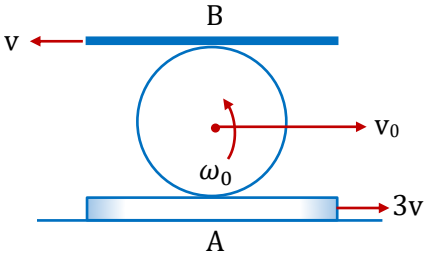
(B) $V_C = V + R\omega$

(C) $a_A = \sqrt{(a - R\alpha)^2 + (\omega^2 R)^2}$

(D) $a_D = \sqrt{(a + \omega^2 R)^2 + (R\alpha)^2}$

PRM162

9. The disc of radius r is confined to roll without slipping at A and B. If the plates have the velocities shown, then



(A) linear velocity $v_0 = v$

(B) angular velocity of disc is $\frac{3v}{2r}$

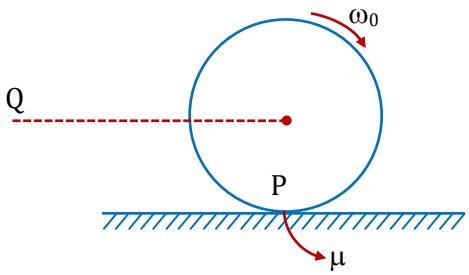
(C) angular velocity of disc is $\frac{2v}{r}$

(D) None of these

PRM168

For More Join: @IITJEE_Advanced

10. A disc of mass 'm' and radius 'R' is rotating with angular speed ω_0 and speed zero as shown and then placed on a rough surface (coefficient of friction = μ) and left. Then



(A) angular momentum of disc is conserved about point Q.

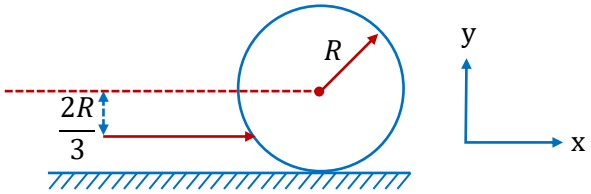
(B) angular momentum of disc is conserved about point P on the ground.

(C) initially the force of friction is μmg leftward.

(D) initially the force of friction is μmg rightward.

PRM195

11. A billiard ball initially at rest is given a sharp blow by a cue stick. The impulse is horizontal and is applied at a distance $2R/3$ below the centreline, as shown in figure. The initial speed of the ball is υ_0, and the coefficient of kinetic friction is μ_k.



(A) Initially kinetic friction acts in $-\hat{\imath}$ direction

(B) Initially kinetic friction acts in $\hat{\imath}$ direction.

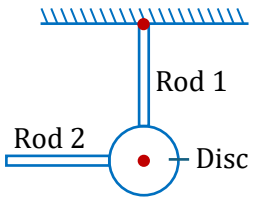
(C) Ball instantaneously starts pure rolling.

(D) Initial angular velocity of ball is $\frac{5v_0}{3R}$

PRM196

12. A disc of radius a is rigidly attached at its circumference by two rods each of length $3a$, two rods are always perpendicular to each other. The combination is suspended vertically from the free end of one rod (as shown in figure). It is swinging in the plane of the disc such that the centre of disc has velocity v. If mass of disc and each rod is m.

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(A) Kinetic energy of the disc $\frac{1}{2}mV^2+\frac{1}{2}\left(\frac{ma^2}{2}\right)\left(\frac{V}{4a}\right)^2$

(B) Kinetic energy of the rod 2 is $\frac{1}{2}mV^2+\frac{1}{2}\times\left(\frac{m(3a)^2}{12}\right)\times\left(\frac{V}{4a}\right)^2$

(C) Kinetic energy of the rod 1 is $\frac{1}{2}\times\left(\frac{m(3a)^2}{3}\right)\times\left(\frac{V}{4a}\right)^2$

(D) Kinetic energy of the system of (disc and both rod) is

$$\frac{1}{2}\left[\left(\frac{m(3a)^2}{3}\right)+\left(\frac{ma^2}{2}+m(4a)^2\right)+\left(\frac{m(3a)^2}{12}+m\frac{89}{4}a^2\right)\right]\left(\frac{V}{4a}\right)^2$$

PRM257

13. A particle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground.

(A) Angular momentum of the particle about O is increasing.

(B) Torque of the gravitational force on the particle about O is decreasing.

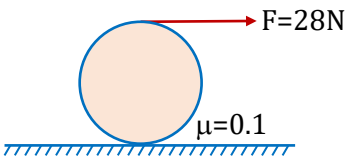
(C) The moment of inertia of the particle about O is decreasing.

(D) The angular velocity of the particle about O is increasing.

PRM134

For More Join: @IITJEE_Advanced

14. A solid sphere of mass 10 *kg* is placed on a rough horizontal surface with coefficient of friction μ = 0.1. If a force of 28 *N* is applied at the top of the sphere then,



(A) friction force will be 10 *N* in forward direction.

(B) friction force will be less than 10*N* in backward direction.

(C) Acceleration of centre of mass will be $\frac{29}{7}\ m/s^2$.

(D) Acceleration of centre of mass will be $3.8 m/s^2$.

PRM203

15. If a cylinder is rolling down the incline with sliding.

(A) after some time, it may start pure rolling

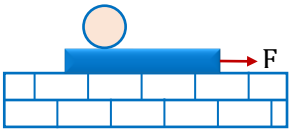
(B) after some time, it will start pure rolling

(C) it may be possible that it will never start pure rolling

(D) none of these

PRM189

16. A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force *F*. If sphere does not slip over the plank. Which of the following is correct.



(A) Acceleration of the centre of sphere is less than that of the plank.

(B) Work done by friction acting on the sphere is equal to its total kinetic energy.

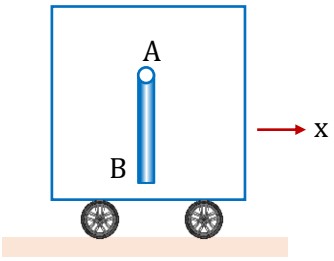
(C) Total kinetic energy of the system is equal to work done by the force *F*.

(D) None of the above

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PRM226

17. A uniform rod *AB* of length ℓ and mass *M* hangs from point *A* at which it is freely hinged in a car moving with velocity v_0. The rod can rotate in vertical plane about the axis at *A*. If the car suddenly stops,



(A) The angular speed ω with which the rod starts rotating is $\frac{3v_0}{2\ell}$.

(B) The minimum value of v_0 so that the rod completes the rotation $\sqrt{\frac{8}{3}g\ell}$.

(C) Loss of energy during the process $\frac{1}{8}Mv_0^2$.

(D) There is no loss of energy.

PRM227

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 18 AND 19

In the calculation of moments of inertia, introductory textbooks often present two theorems, generally called the parallel axis theorem and the perpendicular axis theorem. There is another theorem of this same genre, which is not usually included, but which is interesting and useful. It is

$I_x + I_y + I_z = 2\sum_i m_i r_i^2$

Here, I_x, I_y and I_z are the moments of inertia about three mutually perpendicular intersecting axes, m_1 is the mass of the ith particle and r_i is the distance from the intersection. The proof is simple: Taking the three axes as coordinate axes, we have :

$I_x + I_y + I_z$

$= \sum_i m_i (y_i^2 + z_i^2) + \sum_i m_i (z_i^2 + x_i^2) + \sum_i m_i (x_i^2 + y_i^2)$

$= 2\sum_i m_i (x_i^2 + y_i^2 + z_i^2) = 2\sum_i m_i r_i^2$

One important application is the calculation of the moment of inertia I_d of a uniform thin-walled spherical shell, of mass M and radius R, about a diameter. Taking the centre as the origin of coordinates, we have $I_x = I_y = I_z = I_d$, and $r_i = R$. The theorem gives $3I_d = 2\Sigma_i m_i R^2 = 2MR^2$, where $I_d = \frac{2}{3}MR^2$.

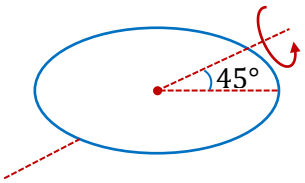
18. Consider a solid cube of mass m and side L. What will be the value of $\sum m_i r_i^2$ for this body when the point of intersection of axes is the centre of the cube :

(A) $\frac{ML^2}{2}$ (B) $\frac{ML^2}{4}$ (C) $\frac{ML^2}{3}$ (D) $\frac{ML^2}{6}$

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PRM106

19. Find the moment of inertia of ring of mass m and radius R about an axis passing through its centre and making an angle 45° with its plane :

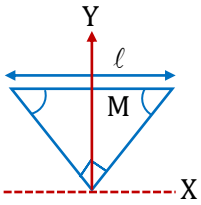


(A) $\frac{MR^2}{4}$ (B) $\frac{MR^2}{2}$ (C) $\frac{3}{4}MR^2$ (D) MR^2

PRM107

PARAGRAPH FOR QUESTION NO. 20 TO 23

The figure shows an isosceles triangular plate of mass M and base L. The angle at the apex is 90°. The apex lies at the origin and the base is parallel to X-axis



For More Join: @IITJEE_Advanced

20. The moment of inertia of the plate about the *z*-axis is :-

(A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{24}$ (C) $\frac{ML^2}{6}$ (D) none of these

PRM112

21. The moment of inertia of the plate about the *x*-axis is :-

(A) $\frac{ML^2}{8}$ (B) $\frac{ML^2}{32}$ (C) $\frac{ML^2}{24}$ (D) $\frac{ML^2}{6}$

PRM113

22. The moment of inertia of the plate about its base parallel to the *x*-axis is :-

(A) $\frac{ML^2}{18}$ (B) $\frac{ML^2}{36}$ (C) $\frac{ML^2}{24}$ (D) none of these

PRM114

23. The moment of inertia of the plate about the *y*-axis is :-

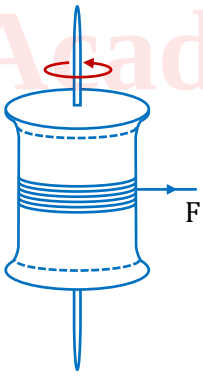
(A) $\frac{ML^2}{6}$ (B) $\frac{ML^2}{8}$ (C) $\frac{ML^2}{24}$ (D) none of these

PRM115

PARAGRAPH FOR QUESTION NO. 24 AND 25

A thread is wrapped around a cylindrical spool of radius 1.5 *cm* whose central axis is fixed on a support. The thread is pulled with a constant acceleration. It is seen that it takes 2 sec to pull 6.28 *cm* of thread starting from rest.

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24. What is the angular velocity of the spool at 1.5 sec ?

(A) π rad/s (B) $\frac{\pi}{2} rad/s$ (C) $\frac{3\pi}{4} rad/s$ (D) $\frac{\pi}{4} rad/s$

PRM261

25. If the thread was pulled with a constant speed so that it takes 2 sec to pull 6.28 *cm* of thread. The angular velocity of spool at 1.5 sec is :

(A) $\frac{4}{3}$ of what was obtained in previous question.

(B) $\frac{2}{3}$ of what was obtained in previous question.

(C) Same as what was obtained in previous question.

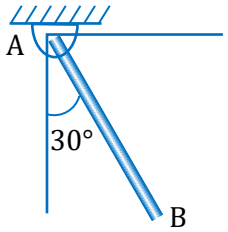
(D) Cannot be predicted.

PRM262

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 26 AND 27

A rod of length ℓ and mass m is hinged at A. It is released from horizontal position from rest. When it comes to position AB, and makes and angle of 30° with vertical



26. Then the angular velocity in this position is :-

(A) $\left(\frac{3\sqrt{3}g}{2\ell}\right)^{1/2}$ (B) $\left(\frac{3g}{2\ell}\right)^{1/2}$ (C) $\left(\frac{9g}{2\ell}\right)^{1/2}$ (D) $\left(\frac{3\sqrt{3}g}{4\ell}\right)^{1/2}$

PRM155

27. Force applied by the hinge on the rod along the rod is :-

(A) $\frac{3mg}{4}$ (B) $\frac{5\sqrt{3}mg}{4}$ (C) $\frac{3\sqrt{3}mg}{4}$ (D) $\frac{\sqrt{3}mg}{2}$

PRM156

PARAGRAPH FOR QUESTION NO. 28 AND 29

A ladder of length L is slipping with its ends against a vertical wall and a horizontal floor. At a certain moment, the speed of the end in contact with the horizontal floor is v and the ladder makes an angle $\alpha = 30°$ with the horizontal.

28. The speed of the ladder's center must be :-

(A) $\frac{2v}{\sqrt{3}}$ (B) $v/2$ (C) v (D) None

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PRM157

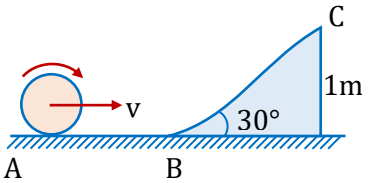
29. If $dv/dt = 0$, then the angular acceleration of the ladder when $\alpha = 45°$ is :-

(A) $2v^2/L^2$ (B) $v^2/2L^2$ (C) $\sqrt{2}\left[\frac{v^2}{L^2}\right]$ (D) None

PRM158

PARAGRAPH FOR QUESTION NO. 30 AND 31

A small sphere of mass 1 kg is rolling without slipping on a rough stationary base with linear speed $v = \sqrt{\frac{200}{7}}\, m/s$. It leaves the inclined plane at point C.



30. Find its linear speed at point C :-

(A) $\sqrt{\frac{100}{7}}\, m/s$ (B) $\sqrt{\frac{50}{7}}\, m/s$ (C) $\sqrt{\frac{100}{35}}\, m/s$ (D) $\sqrt{\frac{200}{35}}\, m/s$

PRM178

31. Find ratio of rotational and translational kinetic energy of the sphere when it strikes the ground after leaving from point C :-

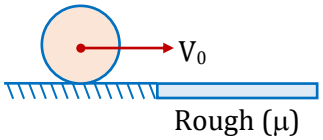
(A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{1}{2}$

PRM179

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 32 TO 35

A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ, as shown in figure.



32. Choose the correct statement(s):

(A) As the ring enters on the rough surface, the limiting friction force acts on it.

(B) The direction of friction is opposite to the direction of motion.

(C) The friction force accelerates the ring in the clockwise sense about its centre of mass.

(D) As the ring enters on the rough surface it starts rolling.

PRM228

33. Choose the correct statement(s):

(A) The momentum of the ring is conserved.

(B) The angular momentum of the ring is conserved about its centre of mass.

(C) The angular momentum of the ring conserved about any point on the horizontal surface in line of friction.

(D) The mechanical energy of the ring is conserved.

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PRM229

34. Choose the correct statement(s):

(A) The ring starts its rolling motion when the centre of mass is stationary.

(B) The ring starts rolling motion when the point of contact becomes stationary.

(C) The time after which the ring starts rolling is $\frac{v_0}{2\mu g}$.

(D) The rolling velocity is $\frac{v_0}{2}$.

PRM230

35. Choose the correct alternative(s):

(A) The linear distance moved by the centre of mass before the ring starts rolling is $\frac{3v_0^2}{8\mu g}$.

(B) The net work done by friction force is $-\frac{3}{8}mv_0^2$.

(C) The loss is kinetic energy of the ring is $\frac{mv_0^2}{4}$.

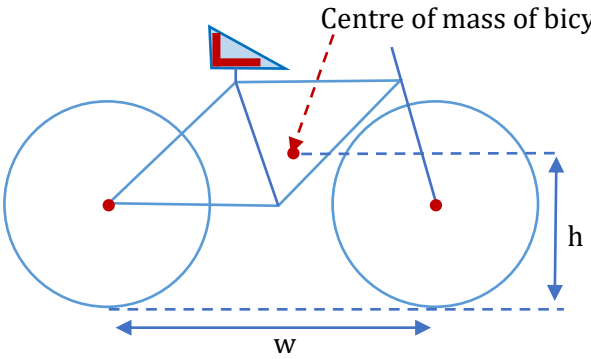
(D) The gain in rotational kinetic energy is $\frac{mv_0^2}{8}$.

PRM231

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 36 TO 38

A simplified model of a bicycle of mass M has two tyres that each comes into contact with the ground at a point. The wheelbase of this bicycle (the distance between the points of contact with the ground) is w, and the center of mass of the bicycle is located midway between the tyres and at height h above the ground. Air resistance may be ignored.



Case-1 (Q. 36) : Assume that cycle is moving rightwards and has acceleration a. Both tyres roll purely.

36. The direction of friction on the tyres is

(A) forward on both
(B) backward on both
(C) forward on front and backward on rear
(D) forward on rear and backward on front.

PRM258

Case-2 (Q. 37 & 38) : The bicycle is moving to the right, but slowing down at a constant rate. The acceleration has a magnitude a. Assume that the coefficient of sliding friction between each tyre and the ground is μ, and that both tyres are skidding (sliding without rotating). Express your answers in terms of w, h, M and g.

Pankaj's Academy JEE

37. What is the maximum value of μ so that both tyres remain in contact with the ground?

(A) $\frac{w}{2h}$
(B) $\frac{h}{2w}$
(C) $\frac{2h}{w}$
(D) $\frac{w}{h}$

PRM259

38. What is the maximum value of a so that both tyres remain in contact with the ground?

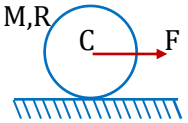
(A) $\frac{wg}{h}$
(B) $\frac{wg}{2h}$
(C) $\frac{hg}{2w}$
(D) $\frac{2hg}{w}$

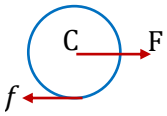
PRM260

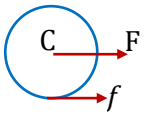
PARAGRAPH FOR QUESTION NO. 39 TO 41

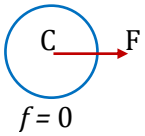
In the following problems, indicate the correct direction of friction force acting on the cylinder, which is pulled on a rough surface by a constant force F.

39. A cylinder of mass M and radius R is pulled horizontally by a force F. The friction force can be given by which of the following diagrams :-



(A) 

(B) 

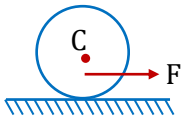
(C) 

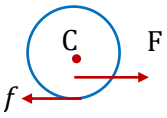
(D) Cannot be interpreted

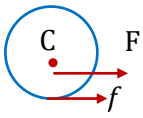
PRM163

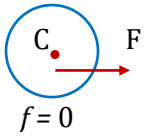
For More Join: @IITJEE_Advanced

40. A cylinder is pulled horizontally by a force F acting at a point below the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams :-



(A) 

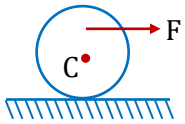
(B) 

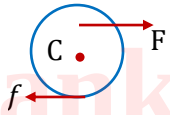
(C) 

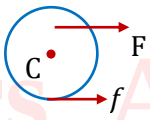
(D) cannot be interpreted

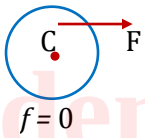
PRM164

41. A cylinder is pulled horizontally by a force F acting at a point above the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams



(A) 

(B) 

(C) 

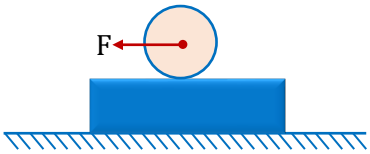
(D) cannot be interpreted

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PRM165

PARAGRAPH FOR QUESTION NO. 42 AND 43

A disc of mass m and radius R is placed over a plank of same mass m. There is sufficient friction between disc and plank to prevent slipping. A force F is applied at the centre of the disc.



42. Acceleration of the plank is :-

(A) $\frac{F}{2m}$ (B) $\frac{3F}{4m}$ (C) $\frac{F}{4m}$ (D) $\frac{3F}{2m}$

PRM198

43. Force of friction between the disc and the plank is :-

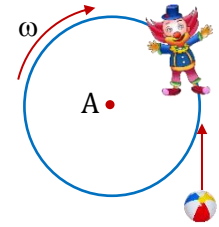
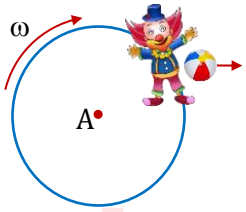
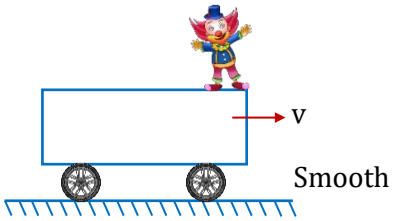
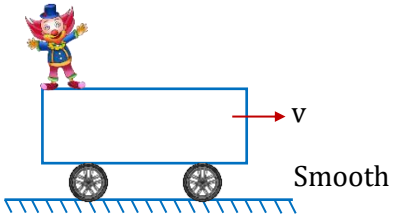
(A) $\frac{F}{2}$ (B) $\frac{F}{4}$ (C) $\frac{F}{3}$ (D) $\frac{2F}{3}$

PRM199

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

44. Column-I depicts various situations where some sudden events are taking place. Column-II describes changes in various parameters of systems immediately after the events taking place in column-I.

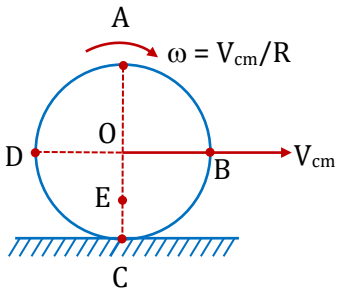
	Column-I		**Column-II**
(A)	Joker is standing on revolving platform and batman throws the ball and joker catches the ball while it was moving horizontally.  Joker, ball and platform is system.	(P)	Linear momentum remains conserved
(B)	Joker throws the ball horizontally and perpendicular to his motion while standing on the revolving platform.  Joker, ball and platform is system.	(Q)	Mechanical energy is conserved
(C)	Joker jumps horizontally towards right from the cart which is moving at speed v on smooth horizontal floor.  Joker and cart is the system	(R)	Mechanical energy increases
(D)	Joker drops himself vertically from the moving cart with no horizontal velocity relative to cart.  Joker and cart is the system	(S) (T)	Mechanical energy decreases. *v* or ω changes

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PRM232

For More Join: @IITJEE_Advanced

45. Consider a body rolling on a horizontal surface as shown in figure:
(Symbols have their usual meaning)



	Column-I		**Column-II**
(A)	Velocity is zero	(P)	Point *A*
(B)	Speed in maximum	(Q)	Point *B*
(C)	0 < speed < v_{cm}	(R)	Point *C*
(D)	1.3v_{cm} < speed < 2 v_{cm}	(S)	Point *D*
		(T)	Point *E*

PRM167

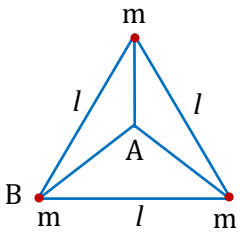
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EXERCISE (S)

Moment of Inertia

1. Three equal masses m are rigidly connected to each other by massless rods of length l forming an equilateral triangle, as shown in the figure. What is the ratio of the moment of inertia of the assembly for an axis through B compared with that for an axis through A (centroid). Both the axis are perpendicular to the plane of triangle.



PRM089

2. Two rods of equal mass m and length l lie along the x axis and y axis with their centres at origin. What is the moment of inertia of the system about the line $x = y$?

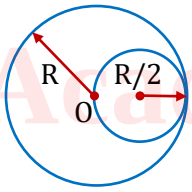
PRM095

3. Find the MI of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is ℓ.

PRM096

4. A hole of radius $R/2$ is cut from a solid sphere of radius R. If the mass of the remaining part of sphere is M, then find moment of inertia of the body about an axis through O.

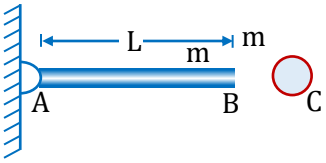
Rankers Academy JEE



PRM108

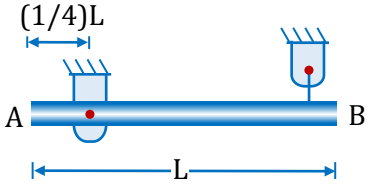
Torque & Condition of Equilibrium

5. A uniform bar AB of mass m and a ball of the same mass are released from rest from the same horizontal position. The bar is hinged at end A. There is gravity downwards. What is the distance of the point from point B that has the same acceleration as that of ball, immediately after release?



PRM149

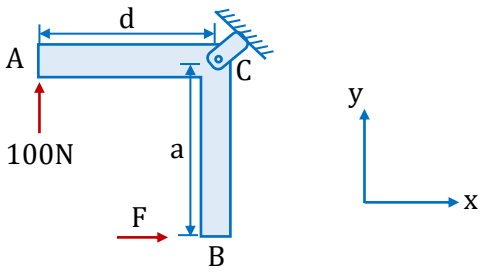
6. A uniform beam of length L and mass m is supported as shown. If the cable at B suddenly breaks, determine; (a) the acceleration of end B, (b) the reaction at the pin support.



PRM150

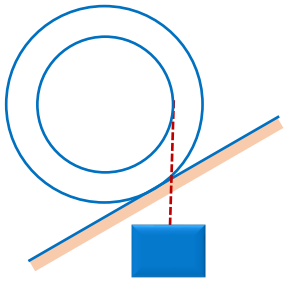
For More Join: @IITJEE_Advanced

7. Find the force F required to keep the system in equilibrium. The dimensions of the system are $d = 0.3\ m$ and $a = 0.2\ m$. Assume the rods to be massless.



PRM118

8. A 3.0 kg bobbin consists of a central cylinder of radius 5.0 cm and two end plates each of radius 6.0 cm. It is placed on a slotted incline, where friction is sufficient to prevent sliding. A block of mass 4.5 kg is suspended from a cord wound around the bobbin and passing through the slot under the incline. If the bobbin is in static equilibrium, what is the angle of tilt of the incline?



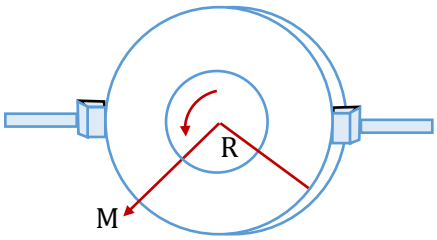
PRM125

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9. A force $\vec{F} = 2\hat{\imath} + 3\hat{\jmath}N$ is applied to an object that is pivoted about a fixed axle aligned along the z coordinate axis. If the force is applied at the point $\vec{r} = 4\hat{\imath} + 5\hat{\jmath}m$, find (a) the magnitude of the net torque about the z-axis and (b) the direction of the torque vector τ.

PRM116

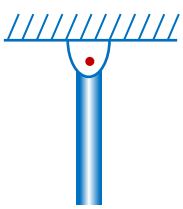
10. A turning moment of 100 Nm is applied to a shaft which passes through a wheel of radius $R = 50cm$. The minimum force with which brakes must be applied to the wheel perpendicular to the surface (see figure) so that the wheel does not rotate is ____ $\times 10^2\ N$. The coefficient of friction equals 0.25.



PRM265

Kinetic Energy in Rotation

11. A rod of mass m *and* length ℓ is hinged at it's upper end. It can rotate in vertical plane. It is given angular velocity ω so that it can complete vertical circle. Find (a) ω (b) Tension at centre of rod at initial moment.

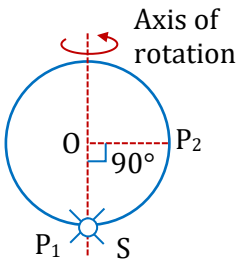


PRM151

For More Join: @IITJEE_Advanced

Fixed Axis Angular Momentum & its Conservation

12. A uniform ring is rotating about vertical axis with angular velocity ω initially. A point insect (S) having the same mass as that of the ring starts walking from the lowest point P_1 and finally reaches the point P_2 (as shown in figure). What is the final angular velocity of the ring?



PRM140

Angular Momentum in Combined Translational & Rotational Motion

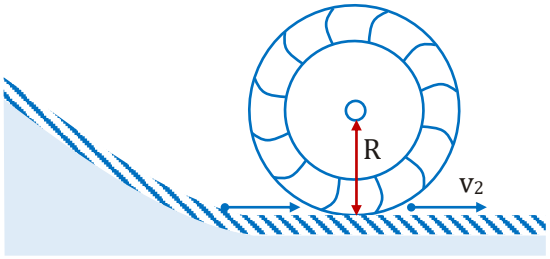
13. A solid sphere of mass m and radius R is placed on a smooth horizontal surface. A sudden blow is given horizontally to the sphere at a height $h = 4R/5$ above the centre line. If I is the impulse of the blow then find:

(a) the minimum time after which the highest point B will touch the ground.

(b) the displacement of the centre of mass during this internal.

PRM191

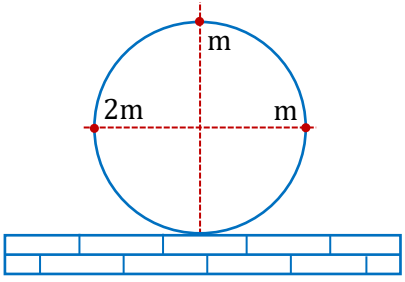
14. Water flowing along an open channel drives an undershot waterwheel of radius 2 m (figure). The water approaches the wheel with a speed of 5.0 m/s and leaves with a speed of 2.5 m/s; the amount of water passing by is 200 kg per second. At what rate does the water deliver angular momentum to the wheel (in kJ) ?



PRM267

Dynamics of Combined Translational & Rotational Motion

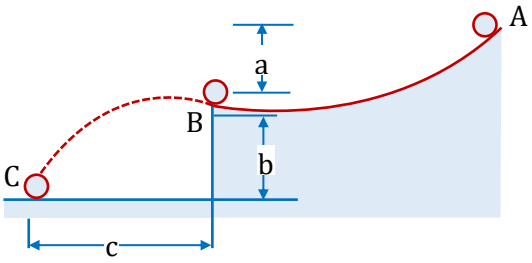
15. A ring of mass m and radius R has three particles attached to the ring as shown in the figure. The centre of the ring has a speed v_0. Find the kinetic energy of the system. (Slipping is absent)



PRM169

For More Join: @IITJEE_Advanced

16. A small sphere of mass m and radius r is released from rest at A and rolls without sliding on the curved surface to point B where it leaves the surface with a horizontal velocity. Knowing that $a = 1.5\ m$ and $b = 1.2\ m$, determine;

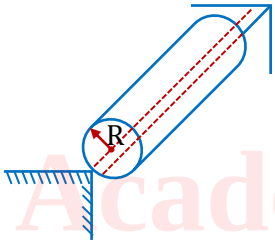


(a) the speed of the sphere as it strikes the ground at C.

(b) the corresponding distance c.

PRM170

17. A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius R is placed horizontally at rest with its length parallel to the edge such that the axis of the cylinder and the edge of the block are in the same vertical plane as shown in figure. there is sufficient friction present at the edge so that a very small displacement causes the cylinder to roll of the edge without slipping. Determine:



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(a) the angle θ_C through which the cylinder rotates before it leaves contact with the edge,

(b) the speed of the centre of mass of the cylinder before leaving contact with the edge, and

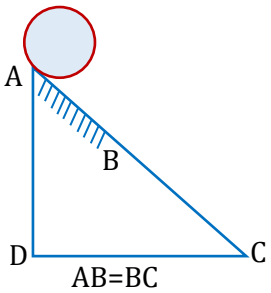
(c) the ratio of the translational to rotational kinetic energies of the cylinder when its centre of mass is in horizontal line with the edge.

PRM176

18. A solid uniform disk of mass m rolls down a fixed inclined plane without slipping with an acceleration a. Find the frictional force on the disk due to surface of the plane.

PRM184

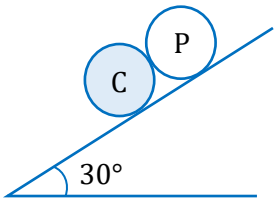
19. Portion AB of the wedge shown in figure is rough and BC is smooth. A solid cylinder rolls without slipping from A to B. Find the ratio of translational kinetic energy to rotational kinetic energy, when the cylinder reaches point C.



PRM186

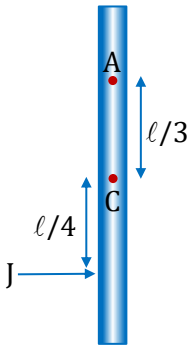
For More Join: @IITJEE_Advanced

20. A solid cylinder *C* and a hollow pipe *P* of same diameter are in contact when they are released from rest as shown in the figure on a long incline plane. Cylinder *C* and pipe *P* roll without slipping. Determine the clear gap (in *m*) between them after 6 seconds.



PRM190

21. A uniform rod of mass *m* and length ℓ is placed in gravity free space and linear impulse *J* is given to the rod at a distance $x = \ell/4$ from centre '*C*' and perpendicular to the rod. Point *A* is at a distance $\ell/3$ from centre as shown in the figure. Then find

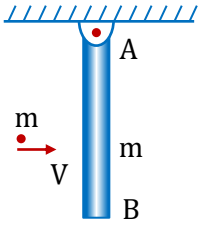


(i) Speed of centre of rod
(ii) Speed of point *A*
(iii) Speed of upper end of rod
(iv) Speed of lower end of rod

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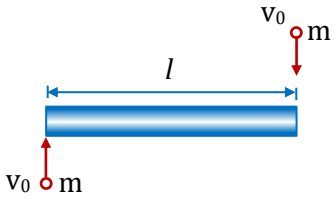
PRM206

22. A uniform rod *AB* of length *L* and mass *m* is suspended freely at *A* and hangs vertically at rest when a particle of same mass *m* is fired horizontally with speed *v* to strike the rod at its mid point. If the particle comes to rest after the impact, then find the impulsive reaction at *A*.



PRM207

23. On a smooth table two particles of mass *m* each, travelling with a velocity v_0 in opposite directions, strike the ends of a rigid massless rod of length *l*, kept perpendicular to their velocity. The particles stick to the rod after the collision. Find the tension in rod during subsequent motion.



PRM208

For More Join: @IITJEE_Advanced

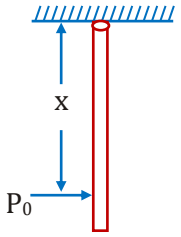
24. A rod AB of mass M and length L is lying on a horizontal frictionless surface. A particle of mass m travelling along the surface hits the end 'A' of the rod with a velocity v_0 in the direction perpendicular to AB. The collision is completely elastic. After the collision the particle comes to rest.

(a) Find the ratio m/M.

(b) A point P on the rod is at rest immediately after the collision. Find the distance AP.

PRM209

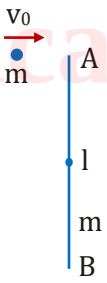
25. A uniform rod of length 0.72 m is pivoted at one end and is hanging as shown. It receives a horizontal impulse at a distance x below the pivot as shown. If impulse applied by the pivot is zero, then find the value of x (in cm).



PRM268

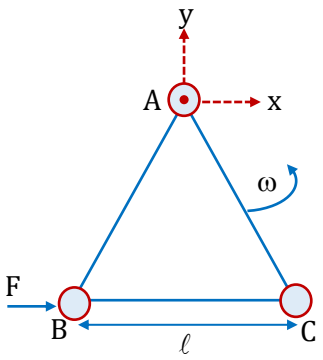
26. A mass m travelling at speed v_0 strikes perpendicularly to a stick of mass m and length $\ell = 1.2m$, which is initially at rest as shown in the figure. The mass collides completely inelastically with the stick at one of its ends, and sticks to it. The location of point from end A (in cm) which is at instantaneous rest just after collision is x_0. Value of $x_0/10$ is

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PRM269

27. Three particles A, B, and C each of mass m, are connected to each other by three massless rigid rods to form a rigid, equilateral triangular body of side ℓ. This body is placed on a horizontal frictionless table (x-y plane) and is hinged to it at the point A, so that it can move without friction about the vertical axis through A (see figure). The body is set into rotational motion on the table about A with a constant angular velocity ω.



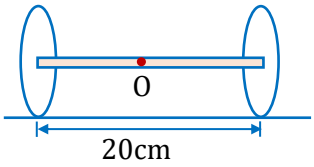
(a) Find the magnitude of the horizontal force exerted by the hinge on the body.

(b) At time T, when the side BC is parallel to the x-axis, a force F is applied on B along BC (as shown). Obtain the x-component and the y-component of the force exerted by the hinge on the body, immediately after time T.

PRM219

For More Join: @IITJEE_Advanced

28. Two thin circular disks of mass 2 *kg* and radius 10 *cm* each are joined by a rigid massless rod of length 20 *cm*. The axis of the rod is along the perpendicular to the planes of the disks through their centre. The object is kept on a truck in such a way that the axis of the object is horizontal and perpendicular to the direction of motion of the truck. Its friction with the floor of the truck is large enough so that the object can roll on the truck without slipping. Take *x*-axis as the direction of motion of the truck and *z*-axis as the vertically upwards direction. If the truck has an acceleration of 9 m/s^2 calculate:



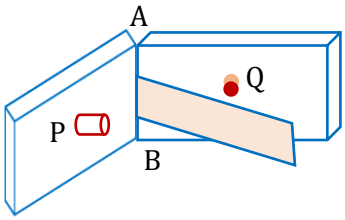
(a) the force of friction on each disk.

(b) the magnitude and the direction of the frictional torque acting on each disk about the centre of mass *O* of the object. Express the torque in the vector form of unit vectors in the *x*-*y* and *z* directions.

PRM221

29. Two heavy metallic plates are joined together at 90° to each other. A laminar sheet of mass 30 *Kg* is hinged at the line *AB* joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to *AB* and passing through its centre of mass is 1.2 $Kg\text{-}m^2$. Two rubber obstacles *P* and *Q* are fixed, one on each metallic plate at a distance 0.5 *m* from the line *AB*. This distance is chosen so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially the laminar sheet hits one of the obstacles with an angular velocity 1 rad/*s* and turns back. If the impulse on the sheet due to each obstacle is 6 *N*–*s*.

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(a) Find the location of the centre of mass of the laminar sheet from *AB*.

(b) At what angular velocity does the laminar sheet come back after the first impact?

(c) After how many impacts, does the laminar sheet come to rest?

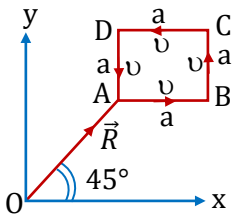
PRM222

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A particle of mass m is moving along the side of a square of side 'a', with a uniform speed υ in the x-y plane as shown in the figure : **[JEE (Main) 2016]**

Which of the following statement is false for the angular momentum $\vec{L}$ about the origin ?



(1) $\vec{L} = \frac{mv}{\sqrt{2}}R\hat{k}$ when the particle is moving from D to A

(2) $\vec{L} = -\frac{mv}{\sqrt{2}}R\hat{k}$ when the particle is moving from A to B

(3) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} - a\right]\hat{k}$ when the particle is moving from C to D

(4) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} + a\right]\hat{k}$ when the particle is moving from B to C

PRM032

2. The moment of inertia of a uniform cylinder of length ℓ and radius R about its perpendicular bisector is I. What is the ratio ℓ/R such that the moment of inertia is minimum? **[JEE (Main) 2017]**

(1) 1 (2) $\frac{3}{\sqrt{2}}$ (3) $\sqrt{\frac{3}{2}}$ (4) $\frac{\sqrt{3}}{2}$

PRM033

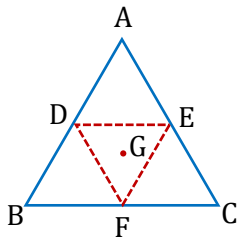
3. To mop-clean a floor, a cleaning machine presses a circular mop of radius R vertically down with a total force F and rotates it with a constant angular speed about its axis. If the force F is distributed uniformly over the mop and if coefficient of friction between the mop and the floor is μ, the torque, applied by the machine on the mop is : **[JEE (Main) 2019]**

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(1) $\frac{2}{3}\mu FR$ (2) $\mu FR/3$ (3) $\mu FR/2$ (4) $\mu FR/6$

PRM034

4. An equilateral triangle ABC is cut from a thin solid sheet of wood. (see figure) D, E and F are the mid-points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0. If the smaller triangle DEF is removed from ABC, the moment of inertia of the remaining figure about the same axis is I. Then: **[JEE (Main) 2019]**

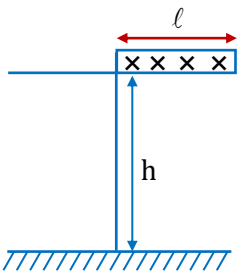


(1) $I = \frac{9}{16}I_0$ (2) $I = \frac{3}{4}I_0$

(3) $I = \frac{I_0}{4}$ (4) $I = \frac{15}{16}I_0$

PRM035

5. A rectangular solid box of length 0.3 m is held horizontally, with one of its sides on the edge of a platform of height 5m. When released, it slips off the table in a very short time τ = 0.01s, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to :- **[JEE (Main) 2019]**



(1) 0.02 (2) 0.28

(3) 0.5 (4) 0.3

PRM036

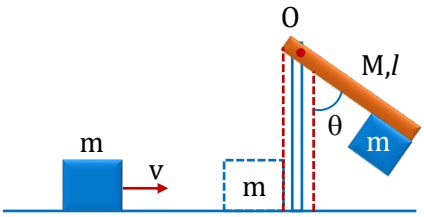
For More Join: @IITJEE_Advanced

6. A stationary horizontal disc is free to rotate about its axis. When a torque is applied on it, its kinetic energy as a function of θ, where θ is the angle by which it has rotated, is given as $k\theta^2$. If its moment of inertia is I then the angular acceleration of the disc is : **[JEE (Main) 2019]**

(1) $\frac{k}{2I}\theta$ (2) $\frac{k}{I}\theta$ (3) $\frac{k}{4I}\theta$ (4) $\frac{2k}{I}\theta$

PRM037

7. A block of mass $m = 1\ kg$ slides with velocity $v = 6\ m/s$ on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass $M = 2\ kg$, and length $l = 1\ m$, the value of θ is approximately : (Take $g = 10\ m/s^2$)

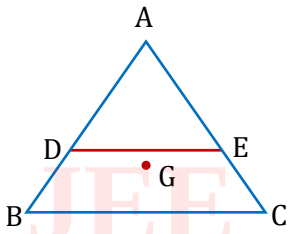


[JEE (Main) 2020]

(1) 69° (2) 63° (3) 55° (4) 49°

PRM039

8. ABC is a plane lamina of the shape of an equilateral triangle. D, E are mid points of AB, AC and G is the centroid of the lamina. Moment of inertia of the lamina about an axis passing through G and perpendicular to the plane ABC is I_0. If part ADE is removed, the moment of inertia of the remaining part about the same axis is $\frac{NI_0}{16}$ where N is an integer. Value of N is _____. **[JEE (Main) 2020]**



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PRM040

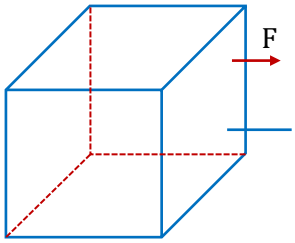
9. Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r) = A + Br$. The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is : **[JEE (Main) 2020]**

(1) $2\pi a^4\left(\frac{A}{4}+\frac{aB}{5}\right)$ (2) $\pi a^4\left(\frac{A}{4}+\frac{aB}{5}\right)$ (3) $2\pi a^4\left(\frac{aA}{4}+\frac{B}{5}\right)$ (4) $2\pi a^4\left(\frac{A}{4}+\frac{B}{5}\right)$

PRM046

10. Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is $\mu = 0.4$, the maximum possible value of $100 \times \frac{b}{a}$ for a box not to topple before moving is _____ .

[JEE (Main) 2020]

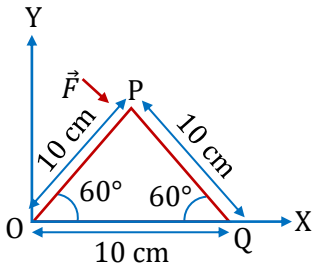


PRM047

For More Join: @IITJEE_Advanced

11. A triangular plate is shown. A force $\vec{F} = 4\hat{i} - 3\hat{j}$ is applied at point *P*. The torque at point *P* with respect to point '*O*' and '*Q*' are:

[JEE (Main) 2021]



(1) – 15 – 20 $\sqrt{3}$, 15 – 20 $\sqrt{3}$

(2) 15 + 20 $\sqrt{3}$, 15 – 20 $\sqrt{3}$

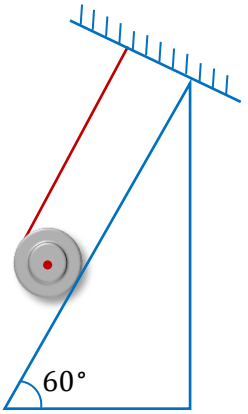
(3) 15 – 20 $\sqrt{3}$, 15 + 20 $\sqrt{3}$

(4) – 15 + 20 $\sqrt{3}$, 15 + 20 $\sqrt{3}$

PRM041

12. A solid cylinder of mass *m* is wrapped with an inextensible light string and, is placed on a rough inclined plane as shown in the figure. The frictional force acting between the cylinder and the inclined plane is :

[JEE (Main) 2021]



[The coefficient of static friction, μ_s, is 0.4]

(1) $\frac{7}{2}mg$

(2) 5 *mg*

(3) $\frac{mg}{5}$

(4) 0

PRM042

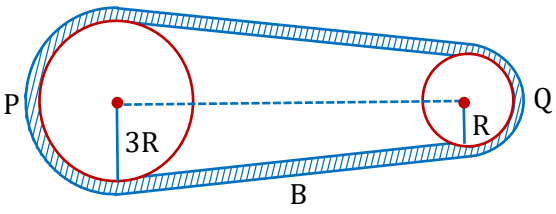
13. A rod of mass *M* and length *L* is lying on a horizontal frictionless surface. A particle of mass '*m*' travelling along the surface hits at one end of the rod with a velocity '*u*' in a direction perpendicular to the rod. The collision is completely elastic. After collision, particle comes to rest. The ratio of masses $\left(\frac{m}{M}\right)$ is $\frac{1}{x}$. The value of '*x*' will be _____.

[JEE (Main) 2021]

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PRM043

14. In the given figure, two wheels *P* and *Q* are connected by a belt *B*. The radius of *P* is three times as that of *Q*. In case of same rotational kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$ will be *x* : 1. The value of *x* will be ____.



[JEE (Main) 2021]

PRM044

15. Two discs have moments of inertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by :

[JEE (Main) 2021]

(1) $\frac{I_1 I_2}{(I_1 + I_2)}(\omega_1 - \omega_2)^2$

(2) $\frac{(I_1 - I_2)^2 \omega_1 \omega_2}{2(I_1 + I_2)}$

(3) $\frac{I_1 I_2}{2(I_1 + I_2)}(\omega_1 - \omega_2)^2$

(4) $\frac{(\omega_1 - \omega_2)^2}{2(I_1 + I_2)}$

PRM045

For More Join: @IITJEE_Advanced

16. A metre scale is balanced on a knife edge at its centre. When two coins, each of mass 10 g are put one on the top of the other at the 10.0 cm mark the scale is found to be balanced at 40.0 cm mark. The mass of the metre scale is found to be $x \times 10^{-2}$ kg. The value of x is **[JEE (Main) 2022]**

PRM050

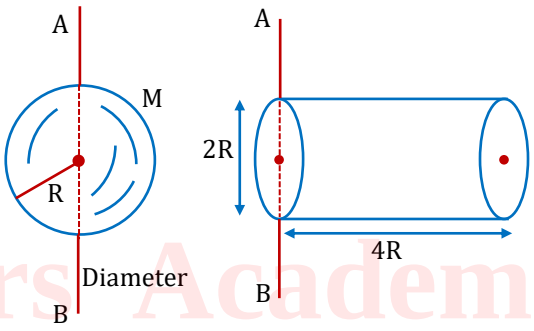
17. A uniform solid cylinder with radius R and length L has moment of inertia I_1, about the axis of cylinder. A concentric solid cylinder of radius $R' = \frac{R}{2}$ and length $L' = \frac{L}{2}$ is caned out of the original cylinder. If I_2 is the moment of inertia of the carved-out portion to the cylinder then $\frac{I_1}{I_2}$ = _____.

(Both I_1 and I_2 are about the axis of the cylinder) **[JEE (Main) 2023]**

PRM272

18. Ratio of radius of gyration of a hollow sphere to that of a solid cylinder of equal mass, for moment of Inertia about their diameter axis AB as shown in figure is $\sqrt{\frac{8}{x}}$. The value of x is:

[JEE (Main) 2024]



Rankers Academy JEE

(1) 34 (2) 17 (3) 67 (4) 51

PRM318

19. The position vectors of two 1 kg particles, (A) and (B), are given by

$\vec{r}_A = (\alpha_1 t^2 \hat{\imath} + \alpha_2 t \hat{\jmath} + \alpha_3 t \hat{k})m$

and $\vec{r}_B = (\beta_1 t \hat{\imath} + \beta_2 t^2 \hat{\jmath} + \beta_3 t \hat{k})m$, respectively ;

(α_1 = 1 m/s^2, α_2 = 3n m/s, α_3 = 2 m/s, β_1 = 2 m/s,

$\beta_2 = -1$ m/s^2, β_3 = 4p m/s), where t is time, n and p are constants, At t = 1s, $|\vec{V}_A| = |\vec{V}_B|$ and velocities $\vec{V}_A$ and $\vec{V}_B$ of the particles are orthogonal to each other. At t = 1s, the magnitude of angular momentum of particle (A) with respect to the position of particle (B) is $\sqrt{L}$ kgm^2s^{-1}. The value of L is ____. **[JEE (Main) 2025]**

PRM392

20. A solid sphere of mass 'm' and radius 'r' is allowed to roll without slipping from the highest point of an inclined plane of length 'L' and makes an angle 30° with the horizontal. The speed of the particle at the bottom of the plane is v_1. If the angle of inclination is increased to 45° while keeping L constant. Then the new speed of the sphere at the bottom of the plane is v_2. The ratio of $v_1^2 : v_2^2$ is

[JEE (Main) 2025]

(1) $1 : \sqrt{2}$ (2) 1 : 3 (3) 1 : 2 (4) $1 : \sqrt{3}$

PRM393

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EXERCISE - JEE (Advanced) PYQ

1. The densities of two solid spheres A and B of the same radii R vary with radial distance r as $\rho_A(r)=k\left(\frac{r}{R}\right)$ and $\rho_B(r)=k\left(\frac{r}{R}\right)^5$, respectively, where k is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B, respectively. If $\frac{I_B}{I_A}=\frac{n}{10}$, the value of n is. **[JEE (Advanced) 2015]**

PRM070

2. A uniform wooden stick of mass 1.6 kg and length ℓ rests in an inclined manner on a smooth, vertical wall of height h ($<\ell$) such that a small portion of the stick extends beyond the wall. The reaction force of the wall on the stick is perpendicular to the stick. The stick makes an angle of 30° with the wall and the bottom of the stick is on a rough floor. The reaction of the wall on the stick is equal in magnitude to the reaction of the floor on the stick. The ratio h/ℓ and the frictional force f at the bottom of the stick are: (g = 10 ms^{-2}) **[JEE (Advanced) 2016]**

(A) $\frac{h}{\ell}=\frac{\sqrt{3}}{16}, f=\frac{16\sqrt{3}}{3}N$ (B) $\frac{h}{\ell}=\frac{3}{16}, f=\frac{16\sqrt{3}}{3}N$

(C) $\frac{h}{\ell}=\frac{3\sqrt{3}}{16}, f=\frac{8\sqrt{3}}{3}N$ (D) $\frac{h}{\ell}=\frac{3\sqrt{3}}{16}, f=\frac{16\sqrt{3}}{3}N$

Rankers Academy JEE

PRM071

3. The position vector of a particle of mass m is given by the following equation $\vec{r}(t)=\alpha t^3\hat{\imath}+\beta t^2\hat{\jmath}$, where $\alpha=\frac{10}{3}ms^{-3}$, β = 5 ms^{-2} and m = 0.1 kg. At t = 1 s, which of the following statement(s) is(are) true about the particle? **[JEE (Advanced) 2016]**

(A) The velocity $\vec{v}$ is given by $\vec{v}=(10\hat{\imath}+10\hat{\jmath})ms^{-1}$.

(B) The angular momentum $\vec{L}$ with respect to the origin is given by $\vec{L}=-\left(\frac{5}{3}\right)\hat{k}\ Nms$.

(C) The force $\vec{F}$ is given by $\vec{F}=(\hat{\imath}+2\hat{\jmath})N$.

(D) The torque $\vec{\tau}$ with respect to the origin is given by $\vec{\tau}=-\left(\frac{20}{3}\right)\hat{k}\ Nm$.

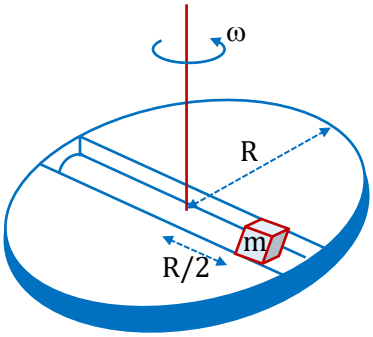
PRM072

PARAGRAPH FOR QUESTION NO. 4 AND 5

A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with a constant angular velocity ω is an example of a non-inertial frame of reference. The relationship between the force $\vec{F}_{\text{rot}}$ experienced by a particle of mass m moving on the rotating disc and the force $\vec{F}_{\text{in}}$ experienced by the particle in an inertial frame of reference is $\vec{F}_{rot}=\vec{F}_{in}+2m(\vec{v}_{rot}\times\vec{\omega})+m(\vec{\omega}\times\vec{r})\times\vec{\omega}$, where $\vec{v}_{rot}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

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Now consider a smooth slot along a diameter of a disc of radius R rotating counter-clockwise with a constant angular speed ω about its vertical axis through its center. We assign a coordinate system with the origin at the centre of the disc, the x-axis along the slot, the y-axis perpendicular to the slot and the z-axis along the rotation axis $(\vec{\omega} = \omega\hat{k})$. A small block of mass m is gently placed in the slot at $\vec{r} = (R/2)\hat{\imath}$ at $t = 0$ and is constrained to move only along the slot.



[JEE (Advanced) 2016]

4. The distance r of the block at time t is :

(A) $\frac{R}{4}(e^{2\omega t}+e^{-2\omega t})$ (B) $\frac{R}{2}\cos 2\omega t$ (C) $\frac{R}{2}\cos \omega t$ (D) $\frac{R}{4}(e^{\omega t}+e^{-\omega t})$

PRM073

5. The net reaction of the disc on the block is :

(A) $-m\omega^2 R\cos\omega t\hat{\jmath} - mg\hat{k}$ (B) $m\omega^2 R\sin\omega t\hat{\jmath} - mg\hat{k}$

(C) $\frac{1}{2}m\omega^2 R(e^{\omega t} - e^{-\omega t})\hat{\jmath} + mg\hat{k}$ (D) $\frac{1}{2}m\omega^2 R(e^{2\omega t} - e^{-2\omega t})\hat{\jmath} + mg\hat{k}$

PRM074

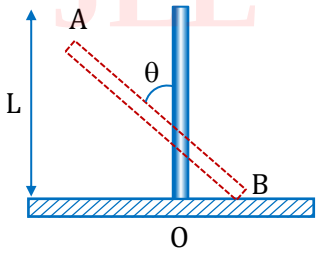
6. A rigid uniform bar AB of length L is slipping from its vertical position on a frictionless floor (as shown in the figure). At some instant of time, the angle made by the bar with the vertical is θ. Which of the following statements about its motion is/are correct ? **[JEE (Advanced) 2017]**

(A) When the bar makes an angle θ with the vertical, the displacement of its midpoint from the initial position is proportional to (1 – cos θ).

(B) The midpoint of the bar will fall vertically downward.

(C) Instantaneous torque about the point in contact with the floor is proportional to sin θ.

(D) The trajectory of the point A is a parabola.



PRM075

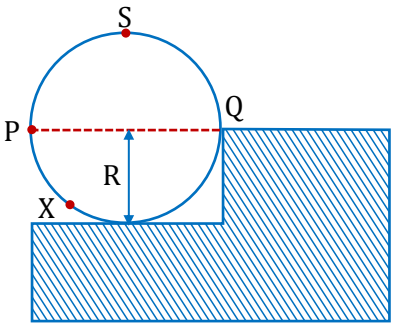
7. A wheel of radius R and mass M is placed at the bottom of a fixed step of height R as shown in the figure. A constant force is continuously applied on the surface of the wheel so that it just climbs the step without slipping. Consider the torque τ about an axis normal to the plane of the paper passing through the point Q. Which of the following options is/are correct ? **[JEE (Advanced) 2017]**

(A) If the force is applied normal to the circumference at point X then τ is constant.

(B) If the force is applied tangentially at point S then $\tau \neq 0$ but the wheel never climbs the step.

(C) If the force is applied normal to the circumference at point P then τ is zero.

(D) If the force is applied at point P tangentially then τ decreases continuously as the wheel climbs.



PRM076

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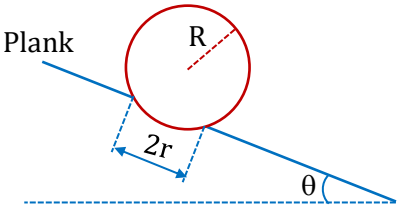
8. The potential energy of a particle of mass m at a distance r from a fixed-point O is given by $V(r) = kr^2/2$, where k is a positive constant of appropriate dimensions. This particle is moving in a circular orbit of radius R about the point O. If v is the speed of the particle and L is the magnitude of its angular momentum about O, which of the following statements is (are) true ?

[JEE (Advanced) 2018]

(A) $v = \sqrt{\frac{k}{2m}}R$ (B) $v = \sqrt{\frac{k}{m}}R$ (C) $L = \sqrt{mk}R^2$ (D) $L = \sqrt{\frac{mk}{2}}R^2$

PRM077

9. A football of radius R is kept on a hole of radius r ($r < R$) made on a plank kept horizontally. One end of the plank is now lifted so that it gets tilted making an angle θ from the horizontal as shown in the figure below. The maximum value of θ so that the football does not start rolling down the plank satisfies (figure is schematic and not drawn to scale)– **[JEE Advanced-2020]**

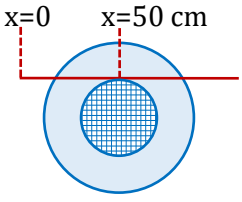


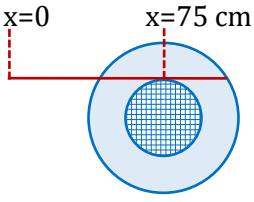
(A) $\sin\theta = \frac{r}{R}$ (B) $\tan\theta = \frac{r}{R}$ (C) $\sin\theta = \frac{r}{2R}$ (D) $\cos\theta = \frac{r}{2R}$

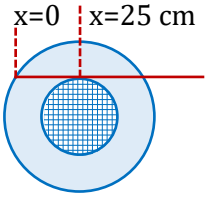
PRM078

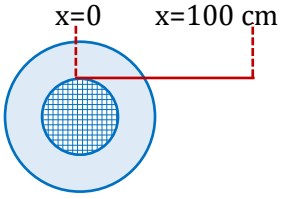
10. A small roller of diameter 20 *cm* has an axle of diameter 10 *cm* (see figure below on the left). It is on a horizontal floor and a meter scale is positioned horizontally on its axle with one edge of the scale on top of the axle (see figure on the right). The scale is now pushed slowly on the axle so that it moves without slipping on the axle, and the roller starts rolling without slipping. After the roller has moved 50 *cm*, the position of the scale will look like (figures are schematic and not drawn to scale)- **[JEE (Advanced) 2020]**



(A)


(B)


(C)


(D)


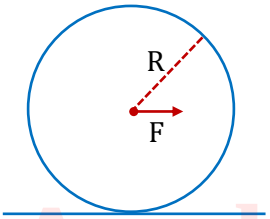
PRM079

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11. Put a uniform meter scale horizontally on your extended index fingers with the left one at 0.00 *cm* and the right one at 90.00 *cm*. When you attempt to move both the fingers slowly towards the center, initially only the left finger slips with respect to the scale and the right finger does not. After some distance, the left finger stops and the right one starts slipping. Then the right finger stops at a distance x_R from the center (50.00 *cm*) of the scale and the left one starts slipping again. This happens because of the difference in the frictional forces on the two fingers. If the coefficients of static and dynamic friction between the fingers and the scale are 0.40 and 0.32, respectively, the value of x_R (in *cm*) is _____. **[JEE (Advanced) 2020]**

PRM080

12. A horizontal force *F* is applied at the center of mass of a cylindrical object of mass m and radius *R*, perpendicular to its axis as shown in the figure. The coefficient of friction between the object and the ground is μ. The center of mass of the object has an acceleration *a*. The acceleration due to gravity is *g*. Given that the object rolls without slipping, which of the following statement(s) is(are) correct? **[JEE (Advanced) 2021]**



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(A) For the same *F*, the value of *a* does not depend on whether the cylinder is solid or hollow.

(B) For a solid cylinder, the maximum possible value of *a* is $2\mu g$.

(C) The magnitude of the frictional force on the object due to the ground is always μmg.

(D) For a thin-walled hollow cylinder, $a = \frac{F}{2m}$.

PRM081

13. A particle of mass $M = 0.2\ kg$ is initially at rest in the *xy*-plane at a point $(x = -l, y = -h)$, where $l = 10\ m$ and $h = 1m$. The particle is accelerated at time $t = 0$ with a constant acceleration $a = 10\ m/s^2$ along the positive *x*-direction. Its angular momentum and torque with respect to the origin, in *SI* units, are represented by $\vec{L}$ and $\vec{\tau}$, respectively. $\hat{\imath}, \hat{\jmath}$ and $\hat{k}$ are unit vectors along the positive *x*, *y* and *z*-directions, respectively. If $\hat{k} = \hat{\imath} \times \hat{\jmath}$ then which of the following statement(*s*) is(are) correct ? **[JEE (Advanced) 2021]**

(A) The particle arrives at the point $(x = l, y = -h)$ at time $t = 2s$

(B) $\vec{\tau} = 2\hat{k}$ when the particle passes through the point $(x = l, y = -h)$

(C) $\vec{L} = 4\hat{k}$ when the particle passes through the point $(x = l, y = -h)$

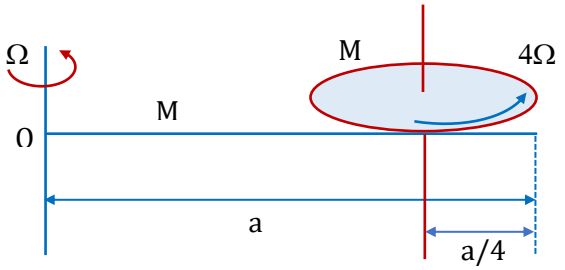
(D) $\vec{\tau} = \hat{k}$ when the particle passes through the point $(x = 0, y = -h)$

PRM082

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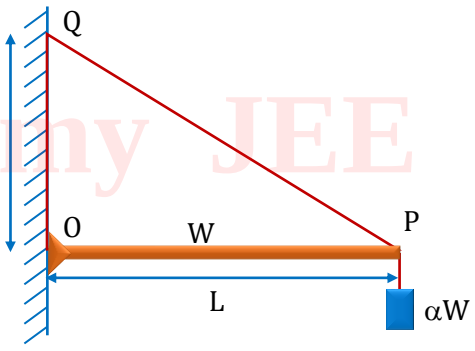
14. A thin rod of mass M and length a is free to rotate in horizontal plane about a fixed vertical axis passing through point O. A thin circular disc of mass M and of radius $a/4$ is pivoted on this rod with its center at a distance $a/4$ from the free end so that it can rotate freely about its vertical axis, as shown in the figure. Assume that both the rod and the disc have uniform density and they remain horizontal during the motion. An outside stationary observer finds the rod rotating with an angular velocity Ω and the disc rotating about its vertical axis with angular velocity 4Ω. The total angular momentum of the system about the point O is $\left(\frac{Ma^2\Omega}{48}\right)n$. The value of n is___.

[JEE (Advanced) 2021]



PRM083

15. One end of a horizontal uniform beam of weight W and length L is hinged on a vertical wall at point O and its other end is supported by a light inextensible rope. The other end of the rope is fixed at point Q, at a height L above the hinge at point O. A block of weight αW is attached at the point P of the beam, as shown in the figure (not to scale). The rope can sustain a maximum tension of $(2\sqrt{2})W$. Which of the following statement(s) is(are) correct ? **[JEE (Advanced) 2021]**



(A) The vertical component of reaction force at O does not depend on α.

(B) The horizontal component of reaction force at O is equal to W for $\alpha = 0.5$.

(C) The tension in the rope is $2W$ for $\alpha = 0.5$.

(D) The rope breaks if $\alpha > 1.5$.

PRM084

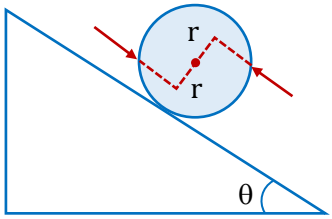
16. At time $t = 0$, a disk of radius 1 m starts to roll without slipping on a horizontal plane with an angular acceleration of $\alpha = \frac{2}{3}$ rad s^{-2}. A small stone is stuck to the disk. At $t = 0$, it is at the contact point of the disk and the plane. Later, at time $t = \sqrt{\pi}s$, the stone detaches itself and flies off tangentially from the disk. The maximum height (in m) reached by the stone measured from the plane is $\frac{1}{2}+\frac{x}{10}$. The value of x is_____. [Take $g = 10\ m\ s^{-2}$.] **[JEE (Advanced) 2022]**

PRM085

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17. A solid sphere of mass 1 kg and radius 1 m rolls without slipping on a fixed inclined plane with an angle of inclination $\theta = 30°$ from the horizontal. Two forces of magnitude 1 N each, parallel to the incline, act on the sphere, both at distance $r = 0.5$ m from the center of the sphere, as shown in the figure. The acceleration of the sphere down the plane is_____ms^{-2}. (Take $g = 10$ $m\ s^{-2}$.)

[JEE (Advanced) 2022]



PRM086

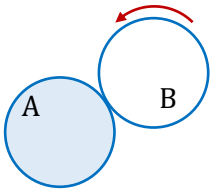
18. A particle of mass 1 kg is subjected to a force which depends on the position as $\vec{F} = -k(x\hat{\imath} + y\hat{\jmath})kgms^{-2}$ with $k = 1$ kgs^{-2}. At time $t = 0$, the particle's position $\vec{r} = \left(\frac{1}{\sqrt{2}}\hat{\imath} + \sqrt{2}\hat{\jmath}\right)m$ and its velocity $\vec{v} = \left(-\sqrt{2}\hat{\imath} + \sqrt{2}\hat{\jmath} + \frac{2}{\pi}\hat{k}\right)ms^{-1}$. Let v_x and v_y denote the x and the y components of the particle's velocity, respectively. **Ignore gravity.** When $z = 0.5$ m, the value of $(x\ v_y - y\ v_x)$ is_____ m^2s^{-1}.

[JEE (Advanced) 2022]

PRM087

19. A flat surface of a thin uniform disk A of radius R is glued to a horizontal table. Another thin uniform disk B of mass M and with the same radius R rolls without slipping on the circumference of A, as shown in the figure. A flat surface of B also lies on the plane of the table. The center of mass of B has fixed angular speed ω about the vertical axis passing through the center of A. The angular momentum of B is $nM\omega R^2$ with respect to the center of A. Which of the following is the value of n?

[JEE (Advanced) 2022]



(A) 2 (B) 5 (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

PRM088

20. A bar of mas $M = 1.00$ kg and length $L = 0.20$ m is lying on a horizontal frictionless surface. One end of the bar is pivoted at a point about which it is free to rotate. A small mass $m = 0.10$ kg is moving on the same horizontal surface with 5.00 ms^{-1} speed on a path perpendicular to the bar. It hits the bar at a distance $L/2$ from the pivoted end and returns back on the same path with speed v. After this elastic collision, the bar rotates with an angular velocity ω. Which of the following statement is correct?

[JEE (Advanced) 2023]

(A) $\omega = 6.98$ rad s^{-1} and $v = 4.30$ ms^{-1} (B) $\omega = 3.75$ rad s^{-1} and $v = 4.30$ ms^{-1}

(C) $\omega = 3.75$ rad s^{-1} and $v = 10.0$ ms^{-1} (D) $\omega = 6.80$ rad s^{-1} and $v = 4.10$ ms^{-1}

PRM273

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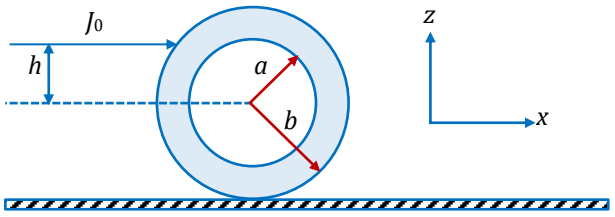
21. An annular disk of mass M, inner radius a and outer radius b is placed on a horizontal surface with coefficient of friction μ, as shown in the figure. At some time, an impulse $J_0\hat{x}$ is applied at a height h above the center of the disk. If $h = h_m$ then the disk rolls without slipping along the x-axis. Which of the following statement(s) is(are) correct ? **[JEE (Advanced) 2023]**

(A) For $\mu \neq 0$ and $a \to 0$, $h_m = b/2$

(B) For $\mu \neq 0$ and $a \to b$, $h_m = b$

(C) For $h = h_m$, the initial angular velocity does **not** depend on the inner radius a.

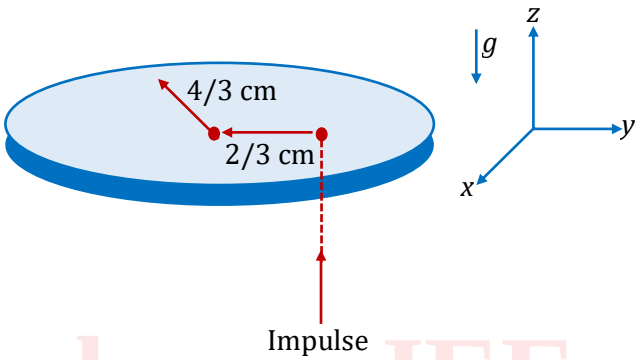
(D) For $\mu = 0$ and $h = 0$, the wheel always slides without rolling.



PRM275

22. A thin circular coin of mass 5 gm and radius $4/3$ cm is initially in a horizontal xy-plane. The coin is tossed vertically up ($+z$ direction) by applying an impulse of $\sqrt{\frac{\pi}{2}} \times 10^{-2} N\text{-}s$ at a distance $2/3$ cm from its center. The coin spins about its diameter and moves along the $+z$ direction. By the time the coin reaches back to its initial position, it completes n rotations. The value of n is ___.

[Given: The acceleration due to gravity g = 10 ms^{-2}]

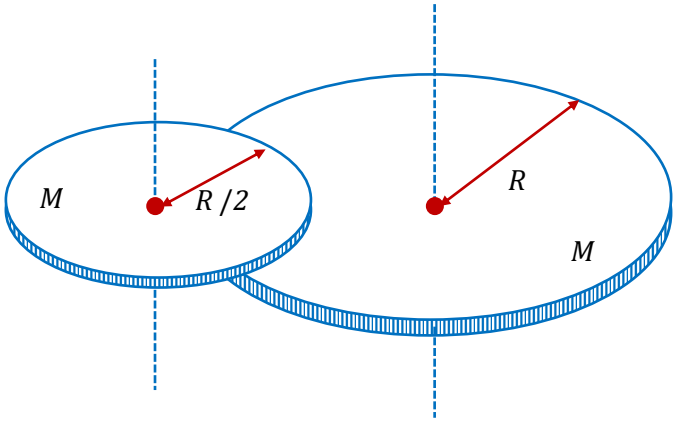


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[JEE (Advanced) 2023]

PRM276

23. A disc of mass M and radius R is free to rotate about its vertical axis as shown in the figure. A battery operated motor of negligible mass is fixed to this disc at a point on its circumference. Another disc of the same mass M and radius $R/2$ is fixed to the motor's thin shaft. Initially, both the discs are at rest. The motor is switched on so that the smaller disc rotates at a uniform angular speed ω. If the angular speed at which the large disc rotates is ω/n, then the value of n is _____.

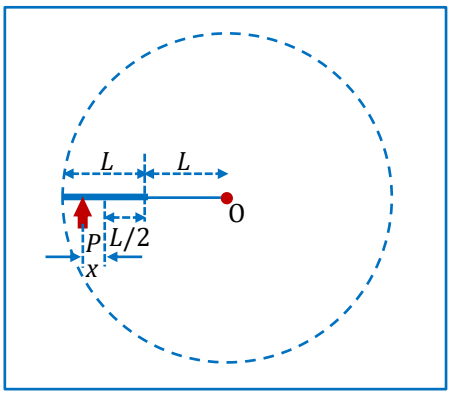


[JEE (Advanced) 2024]

PRM320

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24. A thin uniform rod of length L and certain mass is kept on a frictionless horizontal table with a massless string of length L fixed to one end (top view is shown in the figure). The other end of the string is pivoted to a point O. If a horizontal impulse P is imparted to the rod at a distance $x = L/n$ from the mid-point of the rod (see figure), then the rod and string revolve together around the point O, with the rod remaining aligned with the string. In such a case, the value of n is _____.



[JEE (Advanced) 2024]

PRM321

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	C	D	B	A	D	D	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	B	B	C	D	C	D	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	A	C	C	C	D	A	A	C	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	A	D	B	C	D	B	A	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	D	B	D	C	A	A	B	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	C	A	C	D	B	B	C	A	D
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	B	C	D	B	B	B	D	C	B	B
Que.	71	72	73	74	75	76				
Ans.	A	D	C	A	B	C				

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EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABCD	AC	AD	ABC	AC	ABD	BD	ABCD	AC	BD
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	AD	ACD	ACD	AD	AC	ABC	ABC	B	C	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	C	A	B	A	B	C	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	ABC	C	BCD	ACD	D	A	B	A	A
Que.	41	42	43	44				45		
Ans.	D	C	B	(A-S,T), (B-R), (C-P,R,T), (D-P,Q)				(A-R), (B-P), (C-T), (D-Q,S)		

EXERCISE (S)

1. 2 **2.** $\frac{ml^2}{12}$ **3.** $\frac{a\ell^4}{36}$ **4.** $57/140MR^2$

5. $\frac{L}{3}$ **6.** (a)$\downarrow\frac{9g}{7}$ (b)$\uparrow\frac{4mg}{7}$ **7.** $150(+\hat{\imath})$ **8.** 30°

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9. (a) 2.00 $N{\cdot}m$, (b) $\hat{k}$ **10.** 4

11. (a) $\omega = \sqrt{\frac{6g}{\ell}}$, (b) $T = \frac{11mg}{4}$ **12.** $\frac{\omega}{3}$

13. (a) $t = \frac{\pi Rm}{2I}$; (b) $s = \frac{\pi R}{2}$ **14.** 1 **15.** $6\,mv_0^2$

16. (a) 6.68 m/s (b) 2.27 m **17.** (a) $\theta_C = \cos^{-1}(4/7)$, (b) $v = \sqrt{\frac{4}{7}gR}$, (c) $\frac{K_T}{K_R} = 6$

18. $1/2\ ma$ **19.** 5 **20.** 15

21. (i) $\frac{J}{m}$ (ii) zero (iii) $\frac{J}{2m}$ (iv) $\frac{5}{2}\frac{J}{m}$ **22.** $\frac{mv}{4}$ **23.** $\frac{2mv_0{}^2}{l}$

24. (a) $\frac{m}{M} = \frac{1}{4}$; (b) $x = \frac{2L}{3}$; (c) $\frac{v_0}{2\sqrt{2}}$ **25.** 48 **26.** 8

27. (a) $\sqrt{3}\,ml\omega^2$, (b) $(F_{net})_x = -\frac{F}{4}$, $(F_{net})_y = \sqrt{3}\,ml\omega^2$ **28.** $6N, -0.6\hat{j} \pm 0.6\hat{k}$

29. (a) 0.1 m, (b) 1 rad/s, (c) laminar plate will never come to rest.

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	1	4	3	4	2	11	1	75
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	4	9	3	6	32	3	90	1

Bankers Academy JEE

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	6	D	ABD	D	C	ABC	BC or C	BC	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	25.60	BD	ABC	49	ABD	0.52	2.85 to 2.86	3	B	A
Que.	21	22	23	24						
Ans.	ABCD	30	12	18						

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